STA302 — Methods of Data Analysis

Course Notes

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Week 1: Course Overview and Simple Linear Regression

Learning Objectives

Basic Distinguish population vs. sample quantities, random variables vs. fixed values §; calculate LS coefficient estimates §2; interpret estimated coefficients in context §1.

Adv. Explain the LS process to a broad audience; derive LS estimates for various model specifications; apply SLR in R (`lm()`).

Regression Basics

Definition - Regression

$Y \approx f(X_1, \ldots, X_p)$, $p \geq 1$ predictors. Y = response; X_i = predictors/covariates/features.

- **Simple** ($p = 1$) vs **Multiple** ($p > 1$).
- **Parametric**: f has a fixed parametric form, e.g. $\beta_0 + \beta_1 X$, $\beta_0 + \beta_1 X + \beta_2 X^2$. Else **non-parametric** (kNN, trees, neural nets).

The SLR Model

Definition - SLR Model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \ldots, n, \quad E[\varepsilon_i] = 0 \ \forall i$$

- β_0, β_1: fixed, unknown parameters — *same* for every i.
- ε_i ($= e_i = \epsilon_i$): random, zero-mean — *differs* by i.

Was: $y_i \overset{iid}{\sim} N(\mu, \sigma^2)$. *Now:* $y_i \mid x_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.

Functional ($y_i = \beta_0 + \beta_1 x_i$, no error) vs **statistical** relationship ($y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$); $\varepsilon_i = y_i - (\beta_0 + \beta_1 x_i)$ = residual.

"Linear" = linear in β_0, β_1, not in X.

- OK (still linear): $\beta_0 + \beta_1 x_i^2$, $\beta_0 + \beta_1 \sin x_i$, $\beta_0 + \beta_1 \mathbb{I}(\cdot)$ — free to transform X.
- NOT linear: $\log(\beta_0 + \beta_1 x_i + \varepsilon_i)$, $\beta_0 e^{\beta_1 x_i} + \varepsilon_i$ — *don't transform Y.*

Always condition on X: $E[Y \mid X] = \beta_0 + \beta_1 X$ (abbreviated $E[Y]$, $\mathrm{Var}[Y]$). *$E[Y]$ alone is meaningless — need e.g. $E[Y \mid X = 5]$.*

Notation: sample (x_i, y_i) = realizations of (X, Y); population $Y = \beta_0 + \beta_1 X + \varepsilon$.

Fitted Model & Predictions

Definition - Fitted SLR

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, where $\hat{\beta}_0 \to \beta_0$, $\hat{\beta}_1 \to \beta_1$, $\hat{Y} \to E[Y \mid X]$.

Example 1 Credit Score

Income (X, \$100k)	1.3	1.1	0.8	1.2	1.4	0.9	0.9	1.4	1.2	1.0
Credit Score (Y)	756	728	635	599	760	722	743	726	694	726

`lm(creditScore ~ income)`: $\hat{\beta}_0 = 632.38$, $\hat{\beta}_1 = 68.32$

$$\hat{Y} = 632.38 + 68.32X$$

Interpretation:

- $\hat{\beta}_0$ = avg. Y when $X = 0$ *avg. credit score, \$0 income*
- $\hat{\beta}_1$ = avg. change in Y per unit increase in X *per \$100k income*
- Prediction: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, e.g. $X = 1.05 \Rightarrow \hat{Y} = 704.1$

Validity of predictions — two independent checks:

- **Statistical**: is X inside the training range? (else **extrapolation** — training range was $X \in [0.8, 1.4]$)
- **Subject-matter**: do $X, \hat{Y}$ make sense? (credit score $\notin [300, 900]$, or negative income $\Rightarrow$ invalid)

Least Squares Estimation

$$RSS = \sum_{i=1}^{n} \hat{\varepsilon}_i^2 = \sum_{i=1}^{n} \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)\right)^2 \quad \text{— minimize over } \hat{\beta}_0, \hat{\beta}_1$$

Why square: no +/− cancellation, differentiable, penalizes outliers more; geometrically $\hat{Y}$ = orthogonal projection of Y onto model space.

Identities:

$$\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}, \qquad \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

Proposition 2 LS Estimators.

$$\hat{\beta}_1 = \frac{SP_{xy}}{SS_{xx}} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

SS_{xx} = "sum of squares for x"; SP_{xy} = "sum of products between x, y". No relationship $\Rightarrow \hat{\beta}_1 = 0$, $\hat{\beta}_0 = \bar{y}$.

Derivation. $L(\beta_0, \beta_1) = \sum (y_i - \beta_0 - \beta_1 x_i)^2$.

$$\frac{\partial L}{\partial \beta_0} = 0 \;\Rightarrow\; \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\frac{\partial L}{\partial \beta_1} = 0 \;\Rightarrow\; \sum x_i y_i = \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2$$

Sub. $\hat{\beta}_0$: $\sum x_i y_i - n\bar{x}\bar{y} = \hat{\beta}_1(\sum x_i^2 - n\bar{x}^2) \Rightarrow \hat{\beta}_1$ as above. □

Formulas only valid for 1 untransformed predictor.

Example 3 Deriving for a Different Specification: No Intercept
$y_i = \beta_1 x_i + \varepsilon_i \Rightarrow L(\beta_1) = \sum(y_i - \beta_1 x_i)^2$. $\frac{\partial L}{\partial \beta_1} = -2\sum x_i(y_i - \beta_1 x_i) = 0 \Rightarrow \hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}$. *Same method every time: write L, differentiate, set $= 0$, solve.*

Proposition 4 Properties, if β_0 in model.

$$\sum_{i=1}^{n} \hat{\varepsilon}_i = 0, \qquad \sum_{i=1}^{n} x_i \hat{\varepsilon}_i = 0$$

2nd property: X and residuals uncorrelated (regardless of β_0). Good fit = no pattern in residuals vs. X.

Proof. Sub $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ into both (and $\hat{\beta}_1 = SP_{xy}/SS_{xx}$ into the 2nd). □

Week 2: Multiple Linear Regression (MLR) Models

Learning Objectives	
Basic	Write an MLR model in matrix form §; compute LS estimates via $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ §8; interpret coefficients, indicators, and interactions §.
Adv.	Derive $\hat{\boldsymbol{\beta}}$ via matrix calculus; encode categorical predictors with dummy variables; recognize interaction effects from non-parallel slopes.

Matrices (Review)

Definition - Matrix

A rectangular array of elements in n rows and p columns (*dimension/size* $n \times p$): n = number of rows (1 per *observation*), p = number of columns (1 per *explanatory variable*). *This is exactly* tidy data *format: 1 row per observation, 1 column per variable.*

$$\mathbf{A} = (a_{ij})_{i=1,\ldots,n;\, j=1,\ldots,p} = \begin{pmatrix} a_{11} & \cdots & a_{1p} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{np} \end{pmatrix}$$

Square: $n = p$. **Column vector**: $p = 1$. **Row vector**: $n = 1$.

Definition - Transpose, Symmetric

If $\mathbf{A}$ is $n \times p$, its **transpose** $\mathbf{A}'$ (or $\mathbf{A}^T$) is $p \times n$ (rows $\leftrightarrow$ columns). $\mathbf{A}$ is **symmetric** if $\mathbf{A} = \mathbf{A}'$ (only possible if square).

Example 5 Design Matrix and X′X

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \textit{ vector of responses}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{pmatrix} \textit{ always a column of 1's}$$

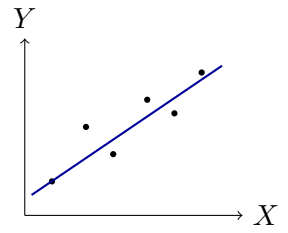
$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{pmatrix} = \begin{pmatrix} 3 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$$

$\mathbf{X}'\mathbf{X}$ is always square and symmetric.

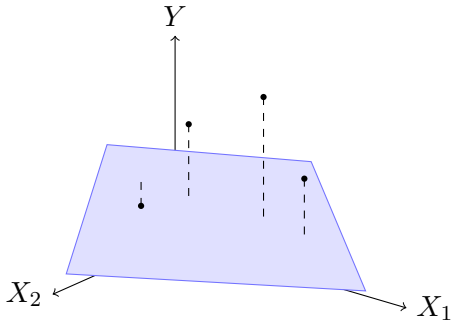
The MLR Model

Definition - Simple vs. Multiple Regression

$p = 1$: **Simple** Regression. $p > 1$: **Multiple** Regression, $f : \mathbb{R}^p \to \mathbb{R}$ — the more general form (SLR is MLR with $p = 1$).



1 predictor: line of best fit



2 predictors: plane in $\mathbb{R}^3$

$p > 2$ predictors: hyperplane in $\mathbb{R}^{p+1}$ (can't visualize directly)

Definition - MLR Model

For individual $i = 1, \ldots, n$:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i, \qquad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$$

where $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$ is treated as fixed/known. Stacking over i: $\boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}}$,

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1' \\ \vdots \\ \mathbf{x}_n' \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix} \sim N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$$

$\sigma^2\mathbf{I}_n$: multivariate normal, diagonal covariance $\mathrm{diag}(\sigma^2, \ldots, \sigma^2)$ *— same σ^2, i.i.d. as in SLR.* $E[Y \mid \mathbf{X} = \mathbf{x}] = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$, $\mathrm{Var}[Y \mid \mathbf{X} = \mathbf{x}] = \sigma^2$ (again abbreviated $E[Y], \mathrm{Var}[Y]$, always conditioning on $\mathbf{X}$).

$\mathbf{x}$ is $(p+1)$-dimensional: 1 intercept + p predictors. (Some sources instead say p entries total, $p-1$ predictors — check convention.)

Remark (Design Matrix Conventions).
- First column is **always** 1's (intercept).
- Must be numeric — categorical predictors need **indicator (dummy) variables**, a.k.a. *one-hot encoding*.
- Each observation must fall in exactly 1 category.

Example 6 Encoding a Categorical Predictor

2 levels (e.g. transmission, automatic/manual): 1 indicator suffices, $X = \mathbb{I}(\text{manual})$.

c **levels** (e.g. colour Blue/Red/Yellow): pick 1 **baseline** (in R: first level alphanumerically), need $c-1$ indicators. $X_2 = \mathbb{I}(\text{Red})$, $X_3 = \mathbb{I}(\text{Yellow})$; both 0 $\Rightarrow$ Blue.

Example 7 Dimension of X

7 numeric predictors, $n = 45$: $\mathbf{X}$ is 45×8 (1 intercept + 7 predictors).

1 continuous predictor + 1 categorical predictor with 10 levels, $n = 100$: $\mathbf{X}$ is 100×11 (1 intercept + 1 continuous + 9 dummy columns). *Want p small relative to n — helps ensure $\mathbf{X}'\mathbf{X}$ is invertible.*

Discrete predictor taking values 1–100: continuous or categorical? Ask: can the levels be ordered? Is the difference between adjacent levels always the same (e.g. $2-1 \stackrel{?}{=} 100-99$)? If yes to both, treat as continuous.

Estimation via Least Squares

Definition - Residual Sum of Squares

$$RSS(\hat{\boldsymbol{\beta}}) = \sum_{i=1}^{n} \hat{\epsilon}_i^2 = \sum_{i=1}^{n} \left(Y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}\right)^2 = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = (\mathbf{Y} - \hat{\mathbf{Y}})'(\mathbf{Y} - \hat{\mathbf{Y}})$$

Same LS procedure as SLR: (1) set up the estimating equation, (2) differentiate w.r.t. $\boldsymbol{\beta}$ and set to $\mathbf{0}$, (3) solve for $\hat{\boldsymbol{\beta}}$, (4) unique minimizer iff $\mathbf{X}'\mathbf{X}$ is invertible (*full rank*) — see *multicollinearity*.

Matrix calculus (Rencher & Schaalje §2.14): for constants $\mathbf{a}$, $\partial(\mathbf{a}'\mathbf{x})/\partial\mathbf{x} = \mathbf{a}$; for symmetric $\mathbf{A}$, $\partial(\mathbf{x}'\mathbf{A}\mathbf{x})/\partial\mathbf{x} = 2\mathbf{A}\mathbf{x}$; and $(\mathbf{A}\mathbf{B})' = \mathbf{B}'\mathbf{A}'$.

Proposition 8 LS Estimator.
Assuming $(\mathbf{X}'\mathbf{X})^{-1}$ exists,

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}}$$

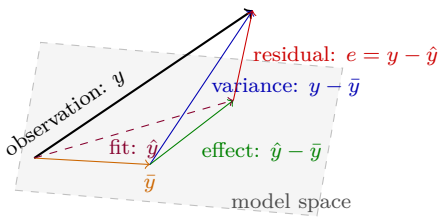
In practice, component matrices are easy to build from sums; only the inverse needs software.

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum x_{i1} & \cdots & \sum x_{ip} \\ \sum x_{i1} & \sum x_{i1}^2 & \cdots & \sum x_{i1}x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_{ip} & \sum x_{i1}x_{ip} & \cdots & \sum x_{ip}^2 \end{pmatrix}, \qquad \mathbf{X}'\mathbf{Y} = \begin{pmatrix} \sum y_i \\ \sum x_{i1}y_i \\ \vdots \\ \sum x_{ip}y_i \end{pmatrix}$$

Definition - Hat Matrix

$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \mathbf{H}\mathbf{Y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the **hat/projection/influence** matrix: *idempotent* ($\mathbf{H}^2 = \mathbf{H}$), and projects $\mathbf{Y}$ onto $\hat{\mathbf{Y}}$ (*model space*).

- Diagonal entries $h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$ are **leverages**: high leverage $\Rightarrow$ *outlier* in X, potential high influence on $\hat{\boldsymbol{\beta}}$ if deleted.
- Residuals: $\hat{\boldsymbol{\epsilon}} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y} = \mathbf{M}\mathbf{Y}$, where $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is the **residual maker** ($\hat{\boldsymbol{\epsilon}} = \mathbf{M}\mathbf{Y}$) or **annihilator** matrix ($\mathbf{M}\mathbf{X} = \mathbf{0}$).



$\hat{y}$ is the orthogonal projection of y onto the model space; $e \perp$ model space.

Application Example

Example 9 `mtcars` Dataset
32 automobiles (1973–74 models); using `mpg` (response), `disp`, `hp`, `wt` (predictors):

$$mpg_i = \beta_0 + \beta_1 disp_i + \beta_2 hp_i + \beta_3 wt_i + \epsilon_i$$

Fitted: $\widehat{mpg} = 37 - 0.009\, disp - 0.03\, hp - 3.8\, wt$. *`disp` not significant ($p = 0.93$); `hp`, `wt` significant ($p = 0.011,\ 0.001$); $R^2 = 0.827$.*

Example 10 Rencher & Schaalje Ex. 7.2 — Worked by Hand
12 observations of Y on X_1, X_2 give $\sum y_i = 90$, $\sum x_{i1} = 52$, $\sum x_{i2} = 102$, $\sum x_{i1}^2 = 296$, $\sum x_{i2}^2 = 1004$, $\sum x_{i1}x_{i2} = 536$, $\sum x_{i1}y_i = 482$, $\sum x_{i2}y_i = 872$, so

$$\mathbf{X'X} = \begin{pmatrix} 12 & 52 & 102 \\ 52 & 296 & 536 \\ 102 & 536 & 1004 \end{pmatrix}, \qquad \mathbf{X'Y} = \begin{pmatrix} 90 \\ 482 \\ 872 \end{pmatrix}$$

Inverting (software) and multiplying:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X'X})^{-1}\mathbf{X'Y} \approx \begin{pmatrix} 5.375 \\ 3.012 \\ -1.285 \end{pmatrix} \Rightarrow\ \hat{y}_i = 5.375 + 3.012x_{i1} - 1.285x_{i2}$$

Example 11 Prediction
Using Ex. 10 at $X_1 = 5$, $X_2 = 9$:

$$\hat{y} = 5.375 + 3.012(5) - 1.285(9) = 8.87$$

Interpretation: the mean response when $X_1 = 5, X_2 = 9$ is 8.87.

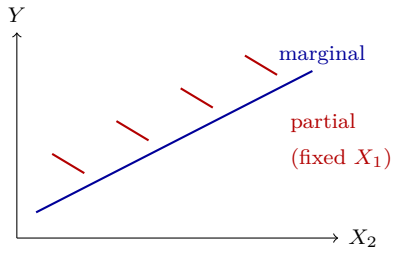
Interpreting Coefficients

Definition - Coefficient Interpretation

$\hat{Y} = \hat{E}[Y \mid \mathbf{X} = \mathbf{x}] = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p = \mathbf{x}'\hat{\boldsymbol{\beta}}$.

- $\hat{\beta}_0$: estimated avg. Y when *all* predictors are 0.
- $\hat{\beta}_j$ $(j \geq 1)$: estimated avg. change in Y per 1-unit increase in x_j, **holding all other predictors fixed**.

Remark (Conditional Nature of MLR). Coefficients are only valid *holding other predictors fixed* — an SLR fit on one predictor alone can differ drastically (even in sign) from its MLR coefficient. In Ex. 10, SLR gives $\hat{y} = 1.86 + 1.30x_1$ and $\hat{y} = 0.86 + 0.78x_2$ (both *positive* slopes), but jointly $\hat{\beta}_2 = -1.285$ (*negative*) — because MLR's $\hat{\beta}_2$ is the effect of X_2 *within* a fixed value of X_1 (a *partial regression*), not *marginally*. *Ignoring a correlated predictor can flip the apparent sign of an effect.*



Marginal slope (blue, ignoring X_1) vs. partial slopes (red, at fixed X_1) — opposite sign here.

Example 12 Interpreting an Indicator Variable
Add `am` ($\mathbb{I}(am = 1)$ = manual) to the `mtcars` model:

$$\widehat{mpg} = \hat{\beta}_0 + \hat{\beta}_1 disp + \hat{\beta}_2 hp + \hat{\beta}_3 wt + \hat{\beta}_4\, \mathbb{I}(am = 1), \qquad \hat{\beta}_4 = 2.16$$

$\hat{\beta}_4$ = estimated avg. difference in *mpg* between manual and automatic transmission, holding `disp`, `hp`, `wt` fixed: the intercept is $\hat{\beta}_0$ when $am = 0$, and $\hat{\beta}_0 + \hat{\beta}_4$ when $am = 1$ — a parallel vertical shift, same slopes.

Definition - Interaction

An **interaction effect** exists when the joint effect of two (or more) predictors differs from their separate effects; include it as a product term $x_j x_k$ (works for any combination of numeric/categorical predictors). *E.g. is the smoking–lung-cancer association the same in rural vs. urban areas? Does a renovation raise price in the suburbs but not downtown? Does drug dosage increase response in females but decrease it in males?*

"Linear" still just means linear in $\boldsymbol{\beta}$: e.g. $Y = \beta_0 + \beta_1 x_1 x_2 + \beta_2 \sin(x_2) + \beta_3 x_3^2$ is a linear model.

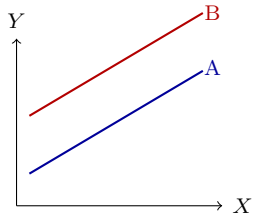
Example 13 Interaction — `mtcars`

$$\widehat{mpg} = \hat{\beta}_0 + \hat{\beta}_1 wt + \hat{\beta}_2 \mathbb{I}(am = 1) + \hat{\beta}_3 (wt \times \mathbb{I}(am = 1))$$

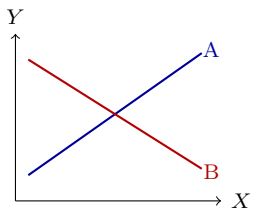
Fitted: $31.42 - 3.79wt + 14.88\,\mathbb{I}(am = 1) - 5.30(wt \times \mathbb{I}(am = 1))$. Slope of `wt` is -3.79 for automatic ($am = 0$); for manual ($am = 1$) it is 5.30 *more negative*, i.e. -9.09 — different intercept *and* slope per group.

Remark (Detecting Interaction Visually). No interaction ⇒ fitted lines/planes for each category are **parallel**. **Non-parallel** slopes (e.g. `mpg` vs. `wt` across `cyl` $\in \{4, 6, 8\}$) suggest an interaction (formal test later). With > 2 predictors, "lines" become planes/hyperplanes.

Plotting caution: don't lead the viewer by extrapolating fitted lines beyond each group's observed range — and don't recode a numeric predictor (e.g. `cyl`) as categorical for a plot like this unless there's a substantive reason to.



No interaction: parallel slopes



Interaction: non-parallel (crossing) slopes

A categorical predictor with $c > 2$ levels still only needs $c - 1$ dummy columns, generated automatically by software (e.g. `factor(cyl)` *for* `cyl` $\in \{4, 6, 8\}$ *creates 2 columns, baseline = smallest level, 4).*

Example 14 Try at Home

Given the data below:

Y	X_1	X_2
-0.2	0.5	0.5
-1.1	0.8	0.7
-0.8	0.6	-0.5
-0.7	-0.2	-0.5

Write out $\mathbf{X}$ (with intercept column) and compute $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ by hand; check against `lm()`.

Week 3: Assumptions of Linear Regression Models

Learning Objectives	
Basic	State the 4 linear regression assumptions and what each means §; use residual plots and QQ plots to check them §.
Adv.	Explain why residuals have non-constant variance and when to standardize; check the two extra conditions needed before residual plots can be trusted in MLR §.

Review: MLR, Indicators and Interactions

Remark (MLR in Matrix Form). $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i$, $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$, i.e. $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$. $\mathbf{Y}$ *= column vector of responses;* $\mathbf{X}$ *= design matrix;* $N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$ *= multivariate normal.*

Example 15 Design Matrix with Interaction — `mtcars`
`lm(mpg ~ wt * factor(cyl))` fits

$$mpg = \beta_0 + \beta_1 wt + \underbrace{\beta_2 \mathbb{I}(cyl = 6) + \beta_3 \mathbb{I}(cyl = 8)}_{\text{main effect for cyl}} + \underbrace{\beta_4 wt \cdot \mathbb{I}(cyl = 6) + \beta_5 wt \cdot \mathbb{I}(cyl = 8)}_{\text{interaction wt}\times\text{cyl}} + \epsilon$$

`model.matrix()` has columns 1, $X_1 = wt$, $X_2 = \mathbb{I}(cyl = 6)$, $X_3 = \mathbb{I}(cyl = 8)$, $X_4 = X_1X_2$, $X_5 = X_1X_3$. *Main effect ⇒ different* intercepts *per group; interaction ⇒ different* slopes *per group.* Fitted: $\hat{\beta}_0 = 39.571$, $\hat{\beta}_1 = -5.647$, $\hat{\beta}_2 = -11.162$, $\hat{\beta}_3 = -15.703$, $\hat{\beta}_4 = 2.867$, $\hat{\beta}_5 = 3.455$ ($R^2 = 0.8616$).

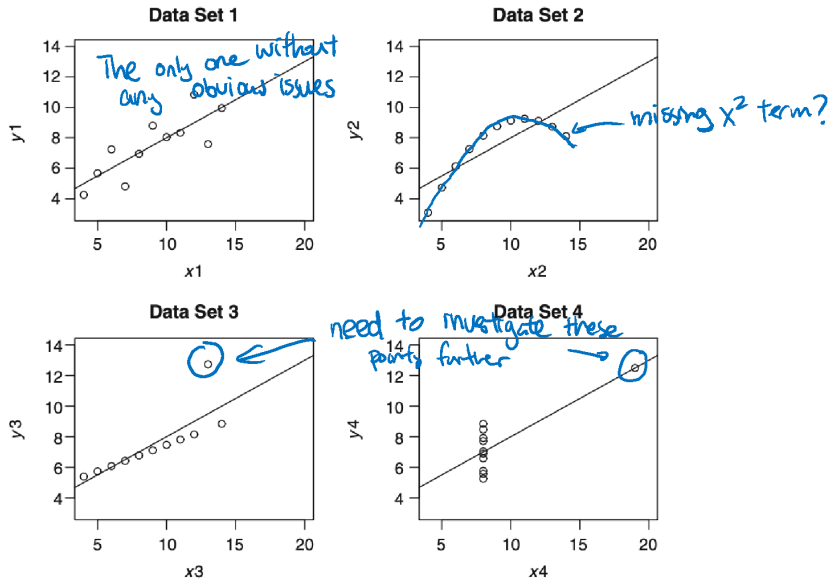
Linear Regression Assumptions

Remark (Why We Need Assumptions). $Y_1, \ldots, Y_n$ are random; treat $\mathbf{X}_1, \ldots, \mathbf{X}_n$ as fixed. So $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ is random, with a **sampling distribution** (like $\hat{\mu}$ in 2nd year). We need assumptions to determine that sampling distribution, to have a valid model fit, and to make statistical inferences.

Example 16 Anscombe's Quartet
Anscombe (1973) built 4 datasets (in R: `anscombe`) with nearly identical summary statistics:

$$\bar{x} = 9, \quad \bar{y} \approx 7.5, \quad \text{Var}(x) = 11, \quad \text{Var}(y) \approx 4.13, \quad \text{corr}(x, y) \approx 0.82, \quad \hat{y} = 3 + 0.5x$$

for *all four*, yet they look completely different when plotted. *Always graph your data! See Sheather §3.1 (pg 45).*



Set 1: the only one without obvious issues. Set 2: missing an x^2 term? Sets 3 & 4: need to investigate the circled points further.

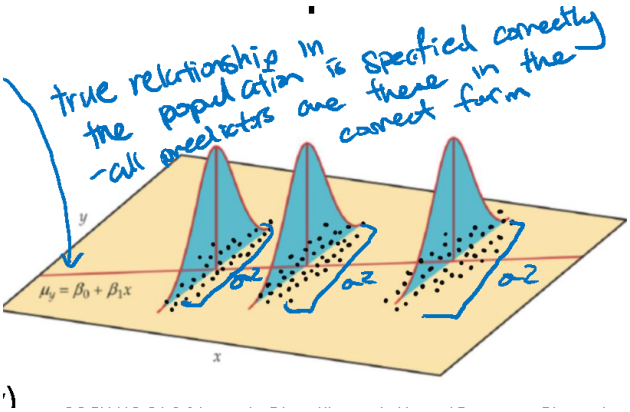
Remark. "All models are wrong, but some are useful" — George Box (1976).

Definition - The 4 Assumptions

Assumptions are about the **population**, not the sample, and we implicitly make them **every** time we fit a model.

1. **Linearity** (a.k.a. **mean zero errors**): $E(\boldsymbol{\epsilon} \mid \mathbf{X}) = \mathbf{0}$, i.e. $E(\mathbf{Y} \mid \mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$
2. **Uncorrelated errors** (a.k.a. independence, no autocorrelation, no serial correlation): $\text{Cov}(\epsilon_i, \epsilon_j) = \text{Cov}(y_i, y_j) = 0$ for $i \neq j$
3. **Constant error variance** (**homoskedasticity**): $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \text{Var}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2\mathbf{I}$, i.e. $\text{Var}(\epsilon_i \mid X) = \sigma^2$
4. **Normal errors**: $\boldsymbol{\epsilon} \mid \mathbf{X} \sim N_n(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$, i.e. $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

The single statement $\boldsymbol{\epsilon} \mid \mathbf{X} \sim N_n(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$ implies all 4 assumptions. Normality is technically not needed for OLS, but is needed for inference on $\boldsymbol{\beta}$.



Each conditional distribution of Y is normal, centred on the line, with the same spread σ^2.

What Each Assumption Means

- **Linearity / mean zero errors**: the true relationship is linear in the *coefficients*, and is exactly $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with
  - no predictors omitted from $\mathbf{X}$ that should be present,
  - no predictors included in $\mathbf{X}$ that should not be present,
  - no predictors in $\mathbf{X}$ in the wrong functional form *(x is transformed if necessary)*.

  Violations: omitting a predictor known to influence the response; fitting a linear model when the truth is $y_i = \log(\beta_0 + \beta_1 x_i + \epsilon_i)$; including x when it should be $x + x^2$. *U-shaped data: $Y = \beta_0 + \beta_1 x$ violates linearity, but $Y = \beta_0 + \beta_1 x + \beta_2 x^2$ is fine — still linear in $\boldsymbol{\beta}$.*
- **Uncorrelated errors**: no data point in the population is related to any other (knowing one tells you nothing about another). Depends on *how samples are taken*. Violations: stock prices, weather data, repeated measurements on the same person, family/social groups.
- **Constant variance**: each conditional distribution has the same *spread*; the only difference between conditional distributions of Y_i is that the mean shifts.
- **Normality**: each conditional distribution has the same (normal) *shape*. Harder to verify in small samples — *but also more important in small samples*. **Not** needed to estimate coefficients by least squares (the normal distribution is never used in that derivation).

Importance of Assumptions

- Linearity ⇒ coefficients are estimated **unbiasedly**
- Uncorrelated errors ⇒ correct **precision** of estimates
- Constant variance ⇒ reasonable estimates of variability for all conditional means
- Normality ⇒ can use properties of normal RVs for **inference** (e.g. CIs)
- Nothing stops us from fitting an invalid model with violated assumptions!

Verifying Assumptions with Residual Plots

Definition - Residuals

$\hat{\boldsymbol{\epsilon}} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}$ are our (sample) estimates of the errors, with

$$E[\hat{\epsilon}_i \mid X] = 0, \qquad \text{Var}[\hat{\epsilon}_i \mid X] = \sigma^2(1 - h_{ii}),$$

where h_{ii} is the i-th diagonal entry of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. *The errors all have the same variance, but the residuals' variances depend on X.* Estimate σ^2 by

$$\hat{\sigma}^2 = \frac{1}{n-(p+1)} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \frac{1}{n-(p+1)} \sum_{i=1}^{n} \hat{\epsilon}_i^2$$

$df = n - p - 1$: n observations minus $p+1$ β's. The sum is the same as in SLR.

Definition - Standardized Residuals

Since residuals don't all have the same variance, standardize:

$$r_i = \frac{\hat{\epsilon}_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}$$

Proposition 17 $0 \le h_{ii} \le 1$.
$\mathbf{H}$ is a projection matrix, so $\mathbf{H} = \mathbf{H}^2$, giving

$$h_{ii} = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2 \;\Rightarrow\; h_{ii} \ge h_{ii}^2 \;\Rightarrow\; 0 \le h_{ii} \le 1.$$

If $h_{ii} \approx 1$ (extremely high leverage), $\hat{\epsilon}_i$ has variance ≈ 0, so it is forced close to 0. *Most h_{ii} will be small (close to 0).* Standardized residual plots are more informative when high-leverage points exist. *Often, regular residual plots are used.*

Remark (Why Residuals Reveal Violations). With population errors $\epsilon_i = y_i - E(y_i \mid x_i)$ and residuals $\hat{\epsilon}_i = y_i - \hat{E}(y_i \mid x_i)$: residuals capture the noise left over after estimating the trend. If $\hat{\boldsymbol{\beta}} \approx \boldsymbol{\beta}$,

$$\underbrace{\hat{\boldsymbol{\epsilon}}}_{\text{observed}} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) + \boldsymbol{\epsilon} \approx \underbrace{\boldsymbol{\epsilon}}_{\text{unobserved}}$$

So violations show up in the residuals. E.g. if the truth is $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \epsilon_i$ but we fit $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1}$:

$$\hat{\epsilon}_i = (\beta_0 - \hat{\beta}_0) + (\beta_1 - \hat{\beta}_1)x_{i1} + \beta_2 x_{i1}^2 + \epsilon_i \approx \beta_2 x_{i1}^2 + \epsilon_i$$

Residuals show what was missed in your specified model.

What Good Residuals Look Like

- Uncorrelated with the fitted values and predictors *(no patterns)*
- Residuals vs. fitted values or any predictor have **constant** mean 0
- **No systematic patterns** — *should look like a random cloud*
- Roughly normally distributed *A funnel (spread growing with x or $\hat{y}$) violates constant variance.*

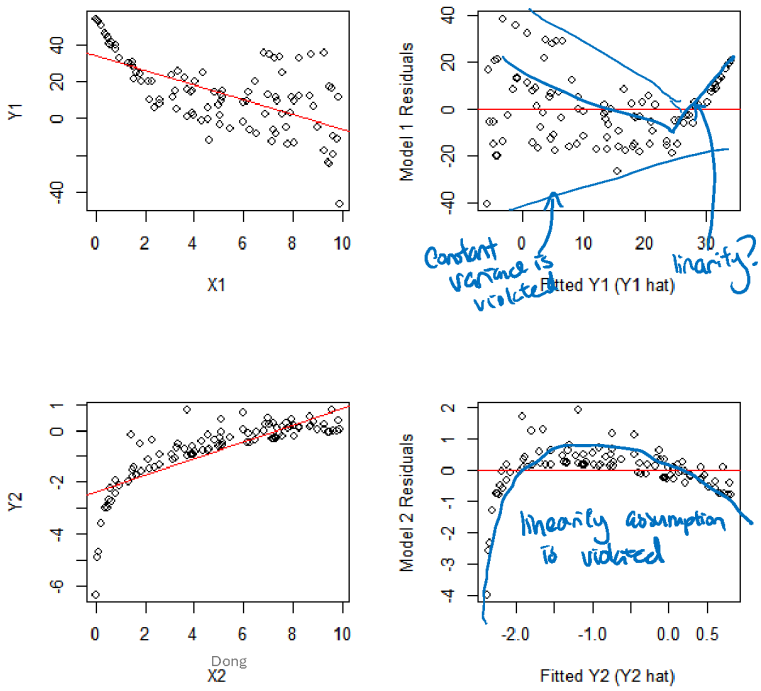
Remark (No Perfect Diagnostic). Diagnostic plots may not capture violations, and may be misleading (especially for small n; more trustworthy for large n). A violation of one assumption may affect the diagnostic plot for *another* assumption.

Plot	Checks
Residuals vs. fitted values (scatter)	linearity, uncorrelated errors, constant variance
Residuals vs. each predictor (scatter)	linearity, uncorrelated errors, constant variance
Histogram of (standardized) residuals	normality, mean zero
Normal QQ plot of residuals	normality

Any of these can use standardized residuals instead.

Example 18 You Try

Scatterplots with line of best fit (left) and residuals vs. fitted (right) for two datasets. What patterns do we see?



(X_1, Y_1): the residuals fan out, so **constant variance is violated**; the curve on the right also suggests a linearity problem. (X_2, Y_2): the residuals follow an arch, so **linearity is violated**.

Definition - Normal QQ Plot

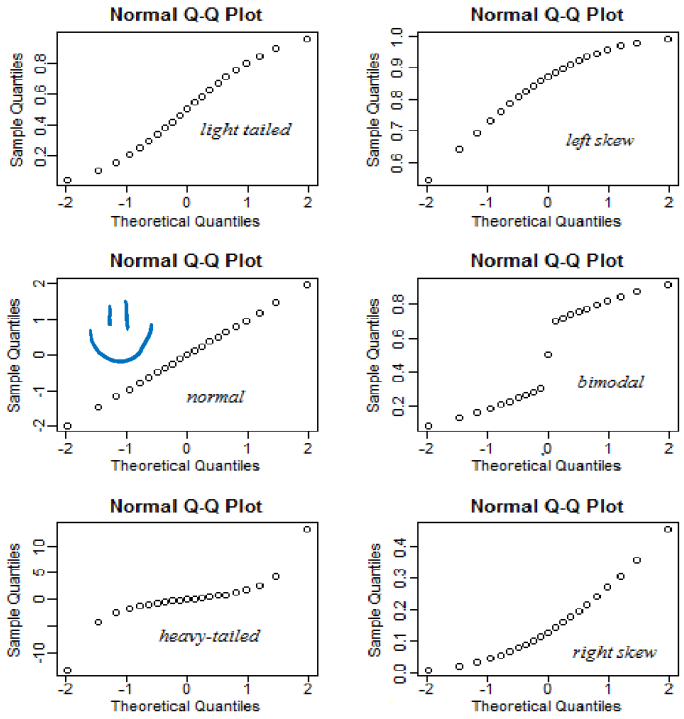
To check normality with a QQ plot of the r_i:

1. Sort the data from smallest to largest; record the percentile each value occupies.
2. Find the standard normal quantile (z-score) at the same percentiles.
3. Plot each data point x_i against its z_i.

If the distribution is close to normal, the points form a **straight line**. In R: `qqnorm(model$residuals)` and `qqline(model$residuals)`.

Remark (Reading QQ Plots). Typical deviations from a straight line:

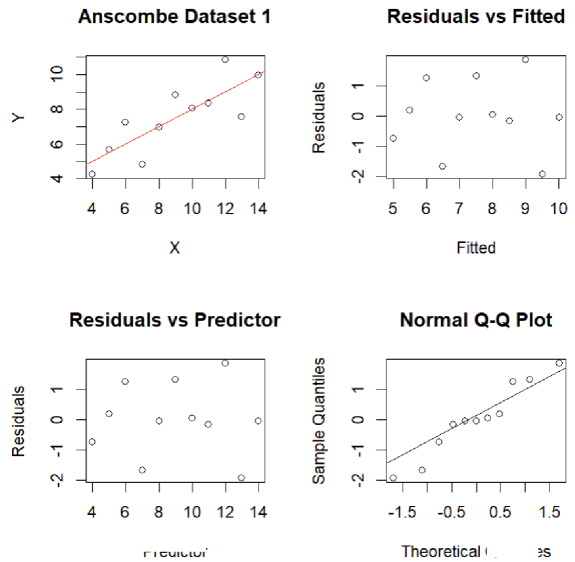
- **Outliers**: odd values far from the rest
- **Curvature at both ends**: long (heavy) or short (light) tails
- **Convex/concave curvature**: asymmetry/skewness
- **Horizontal segments, plateaus, gaps** (e.g. bimodal)



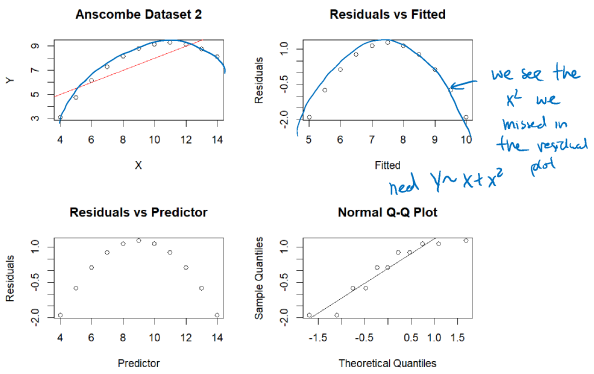
Example 19 Back to Anscombe's Quartet

For each dataset: scatter with fit, residuals vs. fitted, residuals vs. predictor, normal QQ plot.

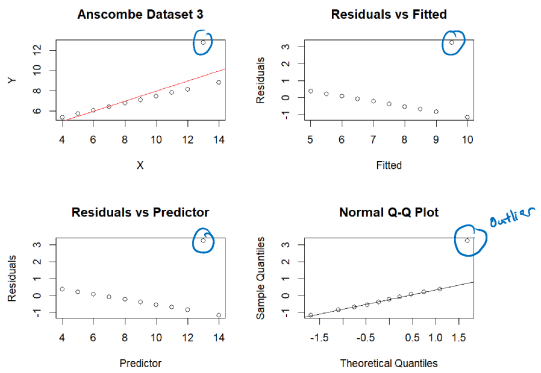
```
model1 <- lm(y1 ~ x1, data = anscombe)
y_hat <- fitted(model1); e_hat <- resid(model1)
par(mfrow = c(2, 2))   # 2x2 grid of plots
plot(anscombe$x1, anscombe$y1); abline(model1, col = "red")
plot(y_hat, e_hat)            # residuals vs fitted
plot(anscombe$x1, e_hat)      # residuals vs predictor
qqnorm(e_hat); qqline(e_hat)
```



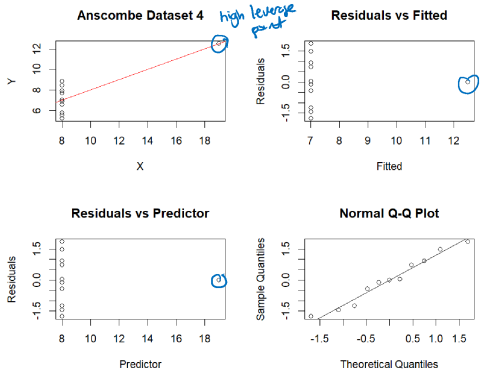
Set 1: random cloud, QQ roughly straight — no issues



Set 2: arch in the residuals



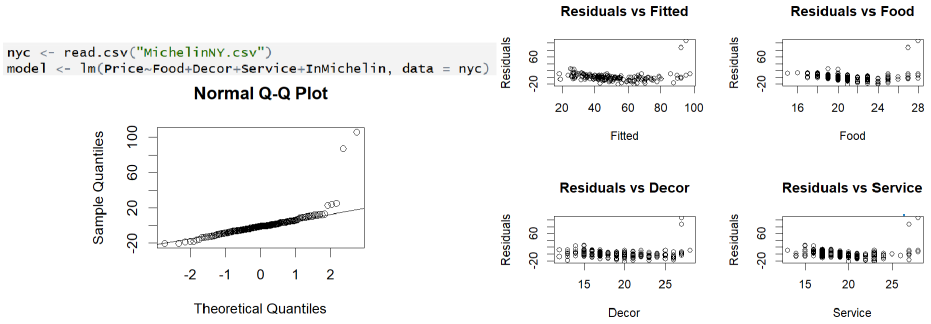
Set 3: one outlier



Set 4: one high-leverage point

Set 2: we see the x^2 we missed in the residual plot — need $Y \sim X + X^2$. Set 3: outlier stands out in every plot. Set 4: the high-leverage point at $x = 19$ has residual ≈ 0 (since $h_{ii} \approx 1$).

Remark (Discrete Predictors). Random vertical "bands" of residuals indicate **no** violation when predictors are discrete (e.g. Michelin NY data, `lm(Price ~ Food + Decor + Service + InMichelin)`).



Explore and Understand the Data

- Do exploratory data analysis *before* fitting a model — plot plot plot!
  - Is most of your data concentrated in a small area? Are the response or predictors skewed? *Doesn't mean something is wrong, but look closer.*
- Existing literature informs knowledge of the true relationship: what are the variables measuring? Are transformations typically done? What predictors mattered in prior studies? How should predictors be included?

Example 20 Think About the Data Collection Process
Which assumptions may be violated?

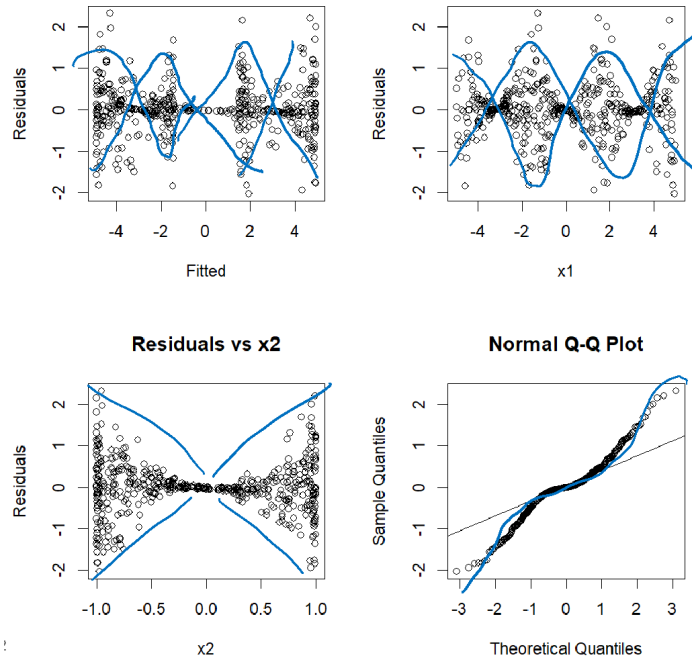
- "Data was collected by voluntary response survey..."
  Linearity: there could be response bias that isn't accounted for in the model.
- "Each variable was measured every day for a month..."
  Uncorrelated errors (consecutive days are related).
- "Neighbourhoods were randomly sampled and all subjects from selected neighbourhoods were included..."
  Uncorrelated errors (people in the same neighbourhood are related).

Some violation is unavoidable (all models are wrong!) — consider the context of the question and the cost of any errors.

Additional Conditions for Multiple Regression Models

The instructor crossed out "Linear" in "Multiple Linear Models": these conditions apply to multiple regression models in general.

Example 21 What's Wrong Here?
The model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ was fit, giving these residual plots:

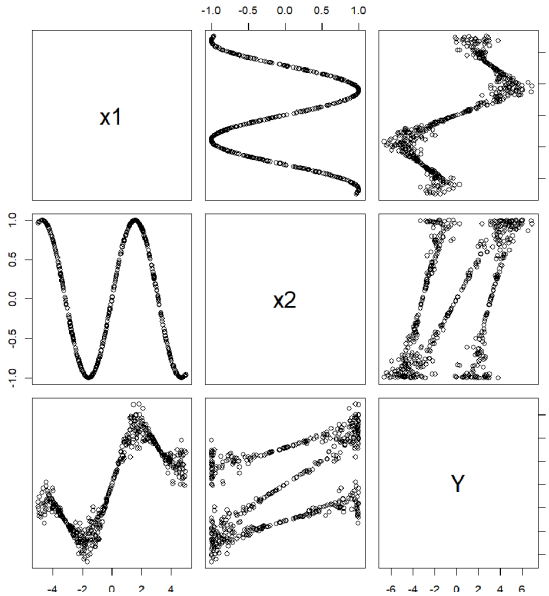


The plots seem to suggest many violations (curved patterns, funnels, non-normal QQ). But the data were generated as

```
x1 = runif(n, -5, 5)
x2 = sin(x1)
Y  = x1 + 3*x2 + rnorm(n, 0, x2^2)
```

The mean is correctly specified and the errors are normal and independent, so **non-constant variance** (sd $= x_2^2$) is the *only* violated assumption.

Example 22 Why Do We See More Issues Than the Funnel?



The pairs plot shows $x_2 = \sin(x_1)$: the predictors have a strongly *non-linear* relationship. That makes the residual plots show patterns that are artifacts of the predictor relationship, not actual extra violations.

Definition - Conditions for Interpreting Residual Plots in MLR

Coefficients are interpreted holding all other predictors fixed (conditioning on them). Before looking at residual plots, check (Li & Duan, 1989; Sheather pg 155):

1. **Conditional mean response condition**: the relationship between the response (or transformed response) and the predictors is linear,

$$E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}) = g(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)$$

2. **Conditional mean predictor condition**: each pair of predictors is linearly related,

$$E(X_i \mid X_j) = \alpha_0 + \alpha_1 X_j$$

These are for interpreting residual plots.

Remark (Why These Conditions?).

- More predictors = more complexity. Residual plots may show patterns that are artifacts of relationships between predictors, not actual violations, and the pattern seen may not be what is actually missing or wrong.
- Recall: violation of one assumption may affect the diagnostic plot for a separate assumption.
- Complex relationships among predictors (or between predictors and response) make residual plots unreliable. Without checking both conditions, we **cannot** use residual plots to identify *specific* violations.

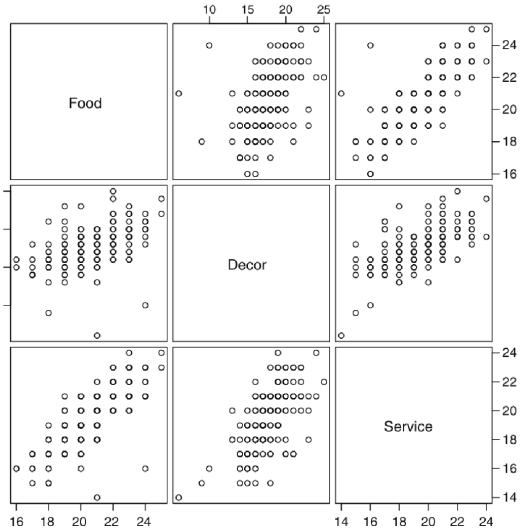
How to Check

- **Response vs. fitted values**: identifies a non-linear relationship. Look for deviations from the line $\hat{Y} = Y$.
- **Pairs plot** (scatterplot matrix): identifies pairwise relationships between predictors. Look for curves and non-linear patterns.

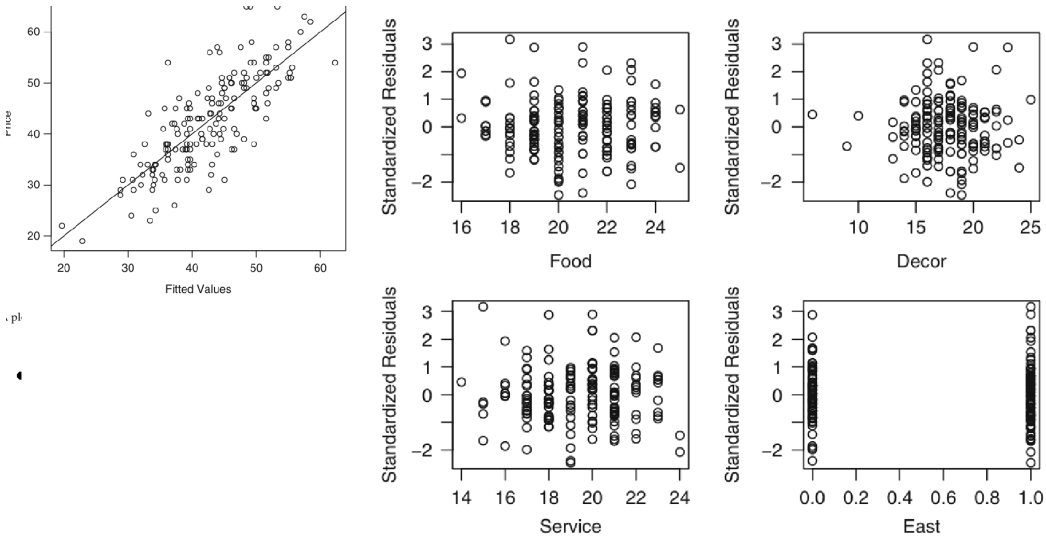
Cook & Weisberg (1999): "Using residuals to guide model development will often result in misdirection, or at best more work than would otherwise be necessary."

Example 23 Both Conditions Hold — NYC Italian Restaurants

3 numeric predictors (Food, Decor, Service) plus East (indicator).



The predictors are (approximately) linearly related ⇒ **conditional mean predictor condition holds**.



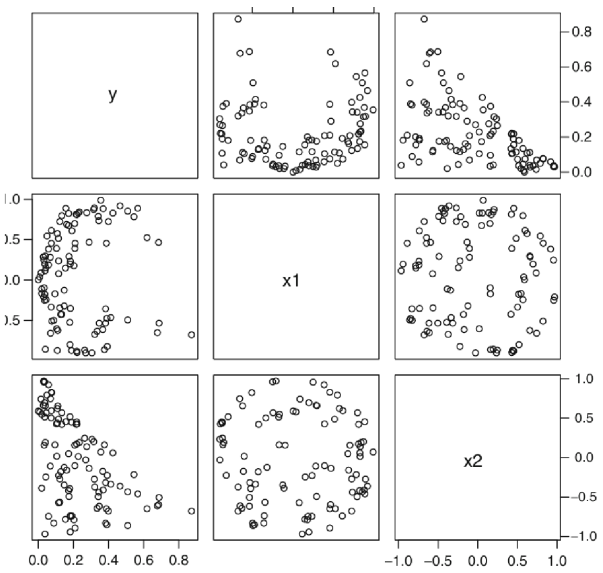
ure 6.2 Plots of standardized residuals against each predictor variable

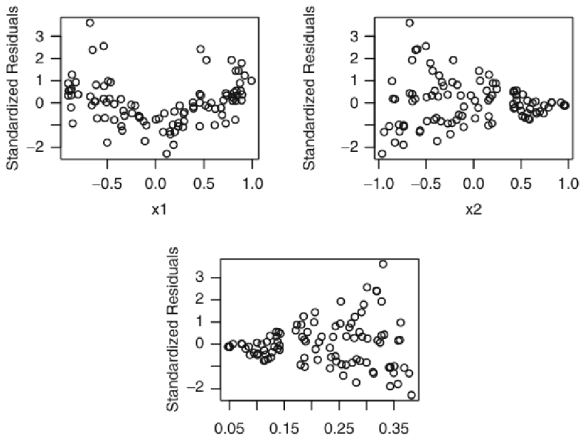
Price vs. fitted values follows the line ⇒ **conditional mean response condition holds** with $g(z) = z$, so the standardized residual plots can be trusted (and look fine).

Example 24 Conditional Mean Response Fails — `alr4::caution`
(Sheather pg 159.) x_1, x_2 generated on the unit circle, errors i.i.d. $N(0, 1)$, and

$$E(Y \mid X) = \frac{|x_1|}{2 + (1.5 + x_2)^2} = \frac{g_1(x_1)}{g_2(x_2)},$$

but the model fit is $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$.





- y has a complicated relationship with x_1 and x_2 $\Rightarrow$ conditional mean response condition **violated**.
- Not enough visual evidence that x_1 vs. x_2 is non-linear $\Rightarrow$ conditional mean predictor condition holds.
- The residual plots look like a constant-variance problem, but we *know* the errors are i.i.d. $N(0, 1)$.

We can't say which part of the model is misspecified from the residual plots. All we can say is that the model fit is invalid.

Summary

- Choose which predictors to use (formal variable selection later)
- Fit the model
- Always plot (extensively!) to check whether the assumptions are valid

Remark (Module Takeaways).

- What are the assumptions of linear regression and what do they mean?
- What tools assess whether each assumption holds, and what do we look for?
- Why check the additional conditions in multiple regression models?
- What aspects of the data, data collection or population suggest an assumption may not hold?

Next week: correcting and mitigating violated assumptions.

Week 4: Mitigating Violated Assumptions

Review: Assumptions and Residual Plots

Remark (The 4 Assumptions (Week 3)). Assumptions are about the **population**, not the sample, and are made implicitly **every** time we fit a model. All four are packed into

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N_n(\underbrace{\mathbf{0}}_{\text{linearity}}, \underbrace{\sigma^2}_{\text{const. var.}} \underbrace{\mathbf{I}}_{\text{uncorr.}}) \quad \text{or} \quad \mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}),$$

with the N itself being the normality assumption.

1. **Linearity** (mean zero errors): $E(\boldsymbol{\epsilon} \mid \mathbf{X}) = \mathbf{0}$, i.e. $E(\mathbf{Y} \mid \mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$
2. **Uncorrelated errors**: $\text{Cov}(\epsilon_i, \epsilon_j) = \text{Cov}(y_i, y_j) = 0$
3. **Constant variance**: $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \text{Var}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2\mathbf{I}$
4. **Normal errors**: $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

1–3 are needed for OLS. Normality is not needed for OLS, but is needed for inference.

Spotting Violations with Residuals

- **Linearity**: any systematic pattern, especially curves or other functions of predictors.
- **Uncorrelated errors**: large clusters of many points, or a pattern in **time** or other sequencing information. *Check how the data was collected.*
- **Constant variance**: any systematic pattern, especially a fanning/funnel pattern with increasing or decreasing spread.
- **Normality**: QQ plot — look for stark deviations/curving/wiggling from the diagonal line *(minimal deviations ok)*.

Introduction to Transformations

Definition - Transformed Model

The main reasons to transform are to overcome **non-linearity** and **non-constant variance**. We fit

$$g_Y(Y) = \beta_0 + \beta_1 g_{X_1}(X_1) + \beta_2 g_{X_2}(X_2) + \cdots + \epsilon$$

where the $g(\cdot)$ are transformations applied to the original data.

- We can transform any subset of the response and predictors.
- We can only interpret the model in terms of the *transformed* predictors/response.

Choosing a Transformation

1. Plot the response vs. the predictors (or residuals) to look for an appropriate transformation.
2. Choose a transformation to try to fix the residual plot.
3. Plot again to see if the issue was fixed.

Drawback: may make interpretation of the estimated coefficients difficult. **Advantage**: often improves prediction and inference.

Definition - Common Transformations

- **Logarithmic**: $g(z) = \log(z)$ *(usually natural log; make sure the variable is strictly positive)*
- **Power**: $g(z) = z^\lambda$ for $\lambda > 0$
- **Inverse power**: $g(z) = 1/z^\lambda$ for $\lambda > 0$

λ is usually a "nice" number. These can be applied to predictors or the response; e.g. from SLR $Y = \beta_0 + \beta_1 X + \epsilon$:

$$Y = \beta_0 + \beta_1 X^2 + \epsilon, \qquad \log(Y) = \beta_0 + \beta_1 X + \epsilon, \qquad \sqrt{Y} = \beta_0 + \beta_1 \exp(X) + \epsilon$$

The last two are no longer linear in Y.

Remark (Transformations for Skewed Data). Instead of X use $X^* = f(X)$ (same for Y); any numerical variable can be transformed.

- **Right skew**: $\log(x - c)$, $\sqrt{x - c}$, etc.
- **Left skew**: x^2, $\sqrt{c - x}$, $\log(c - x)$, etc.
- Check graphically whether it improved.

There are analytic methods to help choose a transformation (Box–Cox, below); these common ones are often the recommended options.

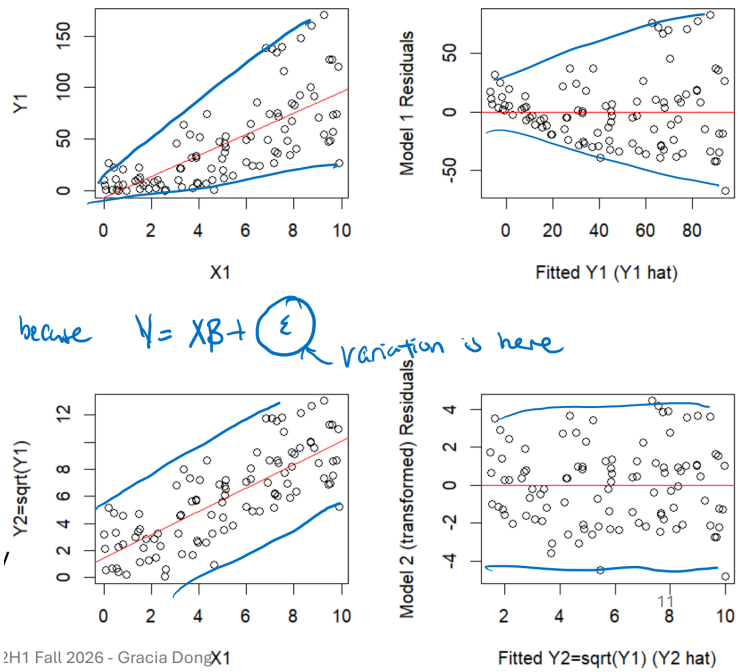
Variance Stabilizing Transformations

Definition - Variance Stabilizing Transformation (VST)

A transformation that specifically targets violation of **constant variance**. It is applied to the **response Y only** — *because in $Y = X\beta + \epsilon$, the variation is in ϵ; there is no randomness associated with X.* Best practice: select transformations that target the violation you observe.

The choice depends on the exact situation and data:

- Identifiable data type (e.g. count data, binary data) or specific variable context
- Experience from similar contexts or data types
- No one transformation works for every model — trial and error



Top: Y_1 vs. X_1 and residuals fan out. Bottom: after $Y_2 = \sqrt{Y_1}$, the residual band is roughly constant.

Proposition 25 Delta Method Approximation.
Let $f(Y)$ be the transformation. First-order Taylor expansion around the mean:

$$f(Y) = f\big(E(Y)\big) + f'\big(E(Y)\big)\big(Y - E(Y)\big) + \cdots$$

Take variance of both sides. $f(E(Y))$, $f'(E(Y))$ and $E(Y)$ are *constants*, so $\mathrm{Var}(f(E(Y))) = 0$ and the covariance term is $\mathrm{Cov}(\text{constant}, \text{random}) = 0$, leaving

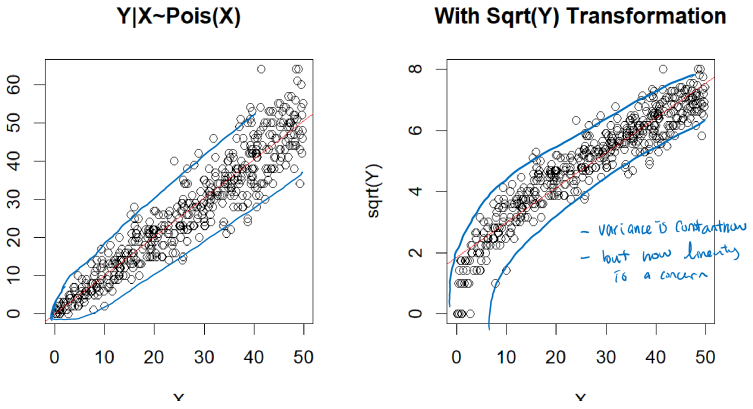
$$\boxed{\mathrm{Var}\left(f(Y)\right) \approx \left[f'(E(Y))\right]^2 \mathrm{Var}(Y)}$$

The correct choice of f cancels out the non-constant part of $\mathrm{Var}(Y)$.

Example 26 Poisson Response
Suppose $Y \mid X \sim \mathrm{Poi}(X)$, so $E(Y \mid X) = \mathrm{Var}(Y \mid X) = X$. *This also violates normality, but focus on non-constant variance for now.* Is $f(Y) = \sqrt{Y}$ variance-stabilizing?
$f(y) = y^{1/2} \Rightarrow f'(y) = \frac{1}{2}y^{-1/2}$, so $f'(E(Y \mid X)) = \frac{1}{2}X^{-1/2}$. Then

$$\mathrm{Var}\left(f(Y) \mid X\right) \approx \left(\tfrac{1}{2}X^{-1/2}\right)^2 X = \tfrac{1}{4},$$

which is **constant**.



After $\sqrt{Y}$ the variance is constant, but now linearity is a concern.

Notes on VSTs

- Only transform Y.
- Can potentially make other assumption violations worse.
- Pick based on what the plots look like and the direction of the funnel:
  - No obvious choice; common ones when variance increases with Y: $\ln(y-c)$, $\sqrt{y-c}$, etc.
  - If variance decreases with Y, flip to $\ln(c-y)$, ...
- Apply transformation, fit model, then plot again.

Proposition 27 Common VSTs.

If $\mathrm{Var}(Y \mid X) \propto$	use $f(Y) \propto$
$E(Y \mid X)$	$\sqrt{Y}$ *(could be $\sqrt{Y-c}$, $\sqrt{c-Y}$, ...)*
$E(Y \mid X)^2$	$\log(Y)$
$E(Y \mid X)^3$	$1/\sqrt{Y}$
$E(Y \mid X)^4$	$1/Y$

It can be hard to tell visually what $\mathrm{Var}(Y \mid X)$ is proportional to — try different transformations and check for improvement.

Exercise 28 Verify the Table.
Verify each row using $\mathrm{Var}(f(Y) \mid X) \approx [f'(E(Y \mid X))]^2 \mathrm{Var}(Y \mid X)$.
Write $\mu = E(Y \mid X)$ and $\mathrm{Var}(Y \mid X) = k\mu^m$. log: $f' = 1/\mu$, so $\mu^{-2} \cdot k\mu^2 = k$. $1/\sqrt{Y}$: $f' = -\frac{1}{2}\mu^{-3/2}$, so $\frac{1}{4}\mu^{-3} \cdot k\mu^3 = k/4$. $1/Y$: $f' = -\mu^{-2}$, so $\mu^{-4} \cdot k\mu^4 = k$. All constant.

Box–Cox Transformations for Normality

Remark (Motivation). Last week we eyeballed transformations for linearity from residual plots and scatterplots. Whatever "signal" the line of best fit misses is most likely a missing predictor (e.g. an arch in residuals vs. Distance ⇒ add Distance^2; a concave $\sqrt{Y}$ vs. X ⇒ try $\sqrt{X}$). Sometimes the needed transformation isn't obvious even from residual vs. predictor plots, so we find the "best" one **analytically**.

Definition - Box–Cox Transformation (Box & Cox, 1964)

Power transformations, chosen to improve **normality** violations. Can be used on the response, on the predictor(s), or on both simultaneously. *Side effect: can help with linearity.* In this class we won't use it exactly as intended due to its complexity.
Further reading: Weisberg §8.1.2–8.1.3 (pg 188); derivation: Appendix A.12 (pg 312).

Remark (Maximum Likelihood). Box–Cox uses ML to estimate the power; the estimate gives the best approximation to normality among all powers. The normal log-likelihood in SLR (assumptions holding) is

$$\log L(\beta_0, \beta_1, \sigma^2 \mid Y) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\underbrace{\sum (y_i - \beta_0 - \beta_1 x_i)^2}_{RSS}$$

For MLR, just include more $\beta_j x_j$ in the last term.

Definition - How Box–Cox Works

Modify the RSS, i.e. fit a new model with a different response:

$$RSS = \sum \left(\Psi_M(Y, \lambda) - \beta_0 - \beta_1 x_i\right)^2$$

$$\Psi_M(\mathbf{Y}, \lambda) = \mathrm{gm}(\mathbf{Y})^{1-\lambda}\, \Psi_S(y_i, \lambda) = \begin{cases} \left(e^{\frac{1}{n}\sum \log y_i}\right)^{1-\lambda} \dfrac{y_i^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \left(e^{\frac{1}{n}\sum \log y_i}\right)^{1-\lambda} \log(y_i), & \lambda = 0 \end{cases}$$

- $\mathrm{gm}(\mathbf{Y}) = e^{\frac{1}{n}\sum \log y_i}$ is the **geometric mean** of Y *(roughly constant if n is large).*
- $\Psi_S(y_i, \lambda) = \frac{y_i^\lambda - 1}{\lambda}$ or $\log(y_i)$ is a **s**caled power transformation.
- $\Psi_M(\mathbf{Y}, \lambda)$ is a **m**odified power transformation applied to $\mathbf{Y}$.

This is very complicated, so for interpretability we instead: get the ML estimate $\hat{\lambda}$ (in R), then just use $\mathbf{Y}^\lambda$ (or $\log \mathbf{Y}$).

Remark (Box–Cox on X, and on X & Y). Predictors: $RSS = \sum \left(y_i - \beta_0 - \beta_1 \Psi_S(X, \lambda)\right)^2$, with $\Psi_S(X, \lambda) = \frac{X^\lambda - 1}{\lambda}$ $(\lambda \neq 0)$ or $\log X$ $(\lambda = 0)$.
Both simultaneously: $RSS = \sum \left(\Psi_M(Y, \lambda_y) - \beta_0 - \beta_1 \Psi_S(X, \lambda_x)\right)^2$, giving ML estimates $(\hat{\lambda}_y, \hat{\lambda}_x)$. We use the simpler Y^{λ_y} and X^{λ_x} (or $\log X$, $\log Y$). For MLR the process is the same, just more λ's to estimate.

Remark (Box–Cox Likelihood — **not testable**, reference only). Issue: $\mathbf{Y} \mid \mathbf{X} \not\sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$; we need Ψ_M so that $\Psi_M(\mathbf{Y} \mid \mathbf{X}) \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$. Modify the log-likelihood and add the Jacobian determinant term (multivariate transformation method, STA237):

$$-\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(\Psi(y_i) - \beta_0 - \beta_1 x_i\right)^2 + n(\lambda - 1)\log\left(\mathrm{gm}(\mathbf{Y})\right)$$

and maximize over λ (in R).

Box–Cox in Practice

- The ML process for λ has no closed form, so we use R.
- The actual Box–Cox transformations are hard to work with, so use **simpler powers**: much easier to **interpret** and apply, still close to normality, and you avoid figuring out what your variable now means.
- Pick something simpler than $\hat{\lambda}$, e.g. $\hat{\lambda} = 0.103 \Rightarrow$ use $\ln(Y)$ (close to 0):

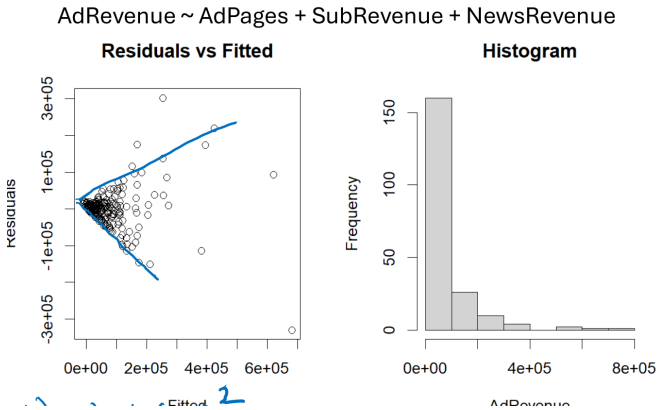
λ	transformation
0.5	square root
0.33	cube root
0.25	fourth root
0	log
-0.5	reciprocal square root
-1	reciprocal (inverse)

- R reports confidence intervals for λ — use them as a range of possibilities.

Application and Interpretation of Transformed Models

Example 29 Magazine Data

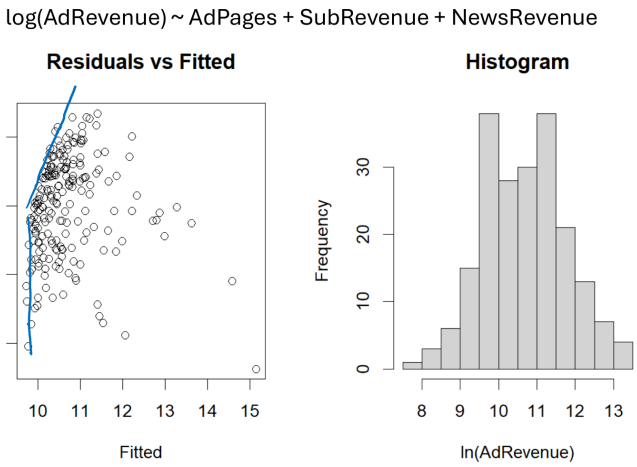
`mag <- read.csv("magazines.csv")`, 204 magazines. Predict $Y =$ `AdRevenue` from $X_1 =$ `AdPages`, $X_2 =$ `SubRevenue`, $X_3 =$ `NewsRevenue`.



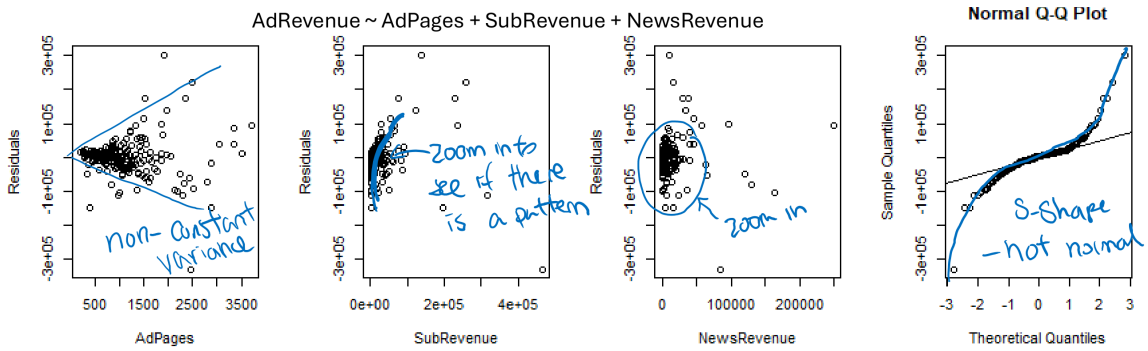
Issues: residuals fan out ($SD(Y) \propto E(Y)$ visually, so $\text{Var}(Y) \propto E(Y)^2$) and `AdRevenue` is strongly right-skewed $\Rightarrow$ **try a** $\log(Y)$ **transformation**.

Example 30 Try $\log($`AdRevenue`$)$

log and $\sqrt{}$ may improve right skew. Some improvement, but other violations may be present (the residuals still have a hard left edge).



More residual plots for the *original* model show many violations:

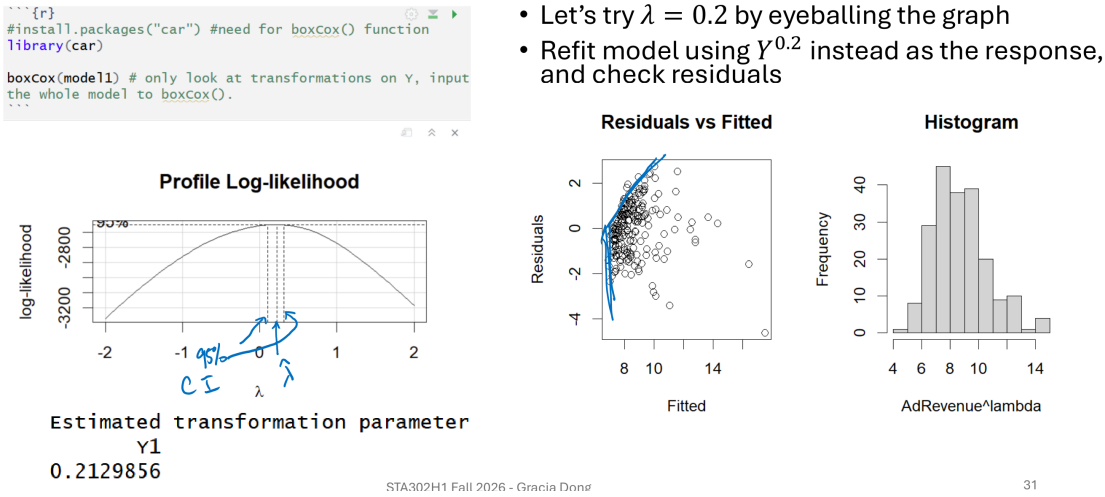


Non-constant variance vs. AdPages; zoom in on SubRevenue/NewsRevenue clusters to see patterns; S-shaped QQ plot ⇒ not normal. So let's try Box–Cox.

Example 31 Box–Cox on Y

Use `boxCox()` from the `car` package:

```
library(car)      # install.packages("car") for boxCox()
boxCox(model1)    # only transformations on Y; input the whole model
```



The profile log-likelihood peaks at $\hat{\lambda} = 0.213$ (dashed lines: $\hat{\lambda}$ and its 95% CI). Eyeballing, try $\lambda = 0.2$: refit with $Y^{0.2}$ as the response and check residuals. *Histogram is more symmetric, but the residuals vs. fitted still have an edge.*

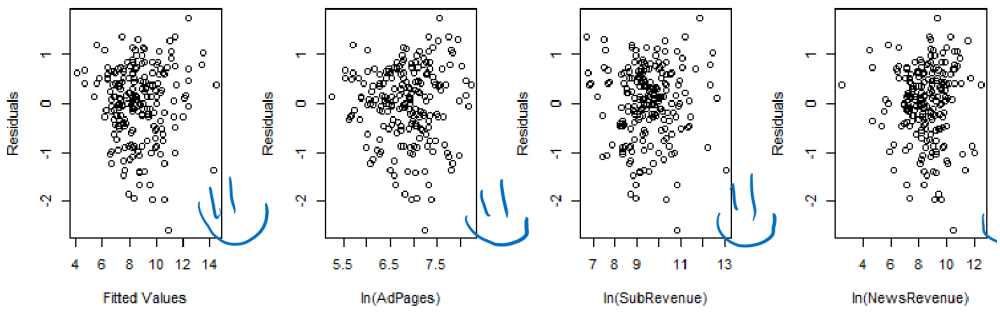
Example 32 Box–Cox on X

Use `powerTransform()` on the predictor columns:

```
p <- powerTransform(cbind(mag[,3:5]))  # all columns you may want to transform
summary(p)
```

	Est Power	Rounded Pwr	Wald Lwr	Wald Upr
AdPages	0.1119	0.00	−0.0869	0.3107
SubRevenue	−0.0084	0.00	−0.0973	0.0804
NewsRevenue	0.0759	0.08	0.0106	0.1412

Ignore the likelihood ratio tests in the output. I would personally round 0.08 again to 0. Refit $Y^{0.2} \sim \log(X_1) + \log(X_2) + \log(X_3)$ ($Y^{0.2}$ from the previous slide) and check residuals:



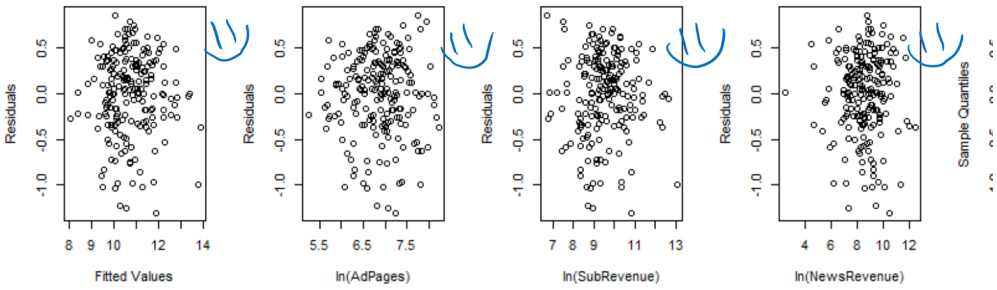
Residual plots look fine; mild C shape in QQ plot — maybe OK, not super concerned.

Example 33 Box–Cox on X and Y Simultaneously
Same function, now including the response column:

```
p <- powerTransform(cbind(mag[,2:5]))  # include X and Y here
summary(p)
```

		Est Power	Rounded Pwr	Wald Lwr	Wald Upr
Y	AdRevenue	0.1071	0.11	0.0299	0.1843
X_1	AdPages	0.0883	0.00	−0.0755	0.2521
X_2	SubRevenue	−0.0153	0.00	−0.0862	0.0557
X_3	NewsRevenue	0.0763	0.08	0.0115	0.1410

Round 0.11 and 0.08 again to 0. Refit $\log(Y) \sim \log(X_1) + \log(X_2) + \log(X_3)$:



Residual plots look fine; mild C shape in QQ plot.

Remark (Interpreting Transformed Models).
- Always interpret using the variables *as they are*: $\log(Y)$, $\log(X)$, etc. If a predictor is squared, say "for a one-unit increase in x^2".
- When the response is transformed, predicted values are conditional means on the **transformed scale**, not the original one. We don't "back-transform" to the original measurement scale.

Example 34 Interpreting the Magazine Model
For $\log(Y) \sim \log(X_1) + \log(X_2) + \log(X_3)$:

- **Intercept**: the expected **log of Revenue** due to Advertising when **log Ad Pages**, **log Subscription Revenue** and **log Newsstand Revenue** are 0.
- **Slope**: for a one unit increase in the **log of Advertising Pages**, the slope is the expected change in **log of Advertising Revenue** for a fixed **log Subscription Revenue** and **log Newsstand Revenue**.

Impact of Violations on Sampling Distributions

Remark (Expected Value Vector and Covariance Matrix). For a random column vector $\mathbf{Y} = (Y_1, \ldots, Y_n)'$: $E[\mathbf{Y}] = (E[Y_1], \ldots, E[Y_n])' = \boldsymbol{\mu}$, and

$$\text{Var}(\mathbf{Y}) = \text{Cov}(\mathbf{Y}) = E\big[(\mathbf{Y} - E[\mathbf{Y}])(\mathbf{Y} - E[\mathbf{Y}])'\big]$$

is $n \times n$, with (i,j)-th entry $\text{Cov}(Y_i, Y_j)$ and i-th diagonal entry $\text{Var}(Y_i)$.

Proposition 35 Linear Transformations.
Let $\mathbf{Y}$ have mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and let $\mathbf{A}$ $(n \times p)$ and $\mathbf{a} \in \mathbb{R}^p$ be constants. Then

$$\begin{aligned} &\text{Var}(\mathbf{Y}) = E(\mathbf{Y}\mathbf{Y}') - \boldsymbol{\mu}\boldsymbol{\mu}' && \textit{cf. } V(Z) = E(Z^2) - E(Z)^2 \\ &E(\mathbf{A}'\mathbf{Y} + \mathbf{a}) = \mathbf{A}'\boldsymbol{\mu} + \mathbf{a} && \textit{cf. } E(cZ+d) = cE(Z) + d \\ &\text{Var}(\mathbf{A}'\mathbf{Y} + \mathbf{a}) = \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A} && \textit{cf. } V(cZ+d) = c^2 V(Z) \end{aligned}$$

If $\mathbf{Y} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\mathbf{A}'\mathbf{Y} \sim N(\mathbf{A}'\boldsymbol{\mu}, \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A})$.

Remark (MLR Setup). Assuming the assumptions hold, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$, i.e. $E[\boldsymbol{\epsilon}] = \mathbf{0}$, $\text{Var}[\boldsymbol{\epsilon}] = \text{diag}(\sigma^2, \ldots, \sigma^2)$. So $\mathbf{Y}$ is multivariate normal too: $\mathbf{Y} \mid \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$.
$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$: $\mathbf{X}$ is fixed and $\mathbf{Y}$ random, so $\hat{\boldsymbol{\beta}}$ is random (a function of $\mathbf{Y}$); $\boldsymbol{\beta}$ is an unknown fixed parameter. Note $\hat{\boldsymbol{\beta}} = \mathbf{A}'\mathbf{Y}$ with $\mathbf{A}' = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Proposition 36 $\hat{\beta}$ is Unbiased.

$$\begin{aligned} E(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) &= E((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{Y} \mid \mathbf{X}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})\boldsymbol{\beta} = \boldsymbol{\beta} \end{aligned}$$

Proposition 37 Variance of $\hat{\beta}$.
Using $\text{Var}(\mathbf{A}'\mathbf{Y}) = \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A}$ with $\boldsymbol{\Sigma} = \text{Var}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2\mathbf{I}$ and $\mathbf{A} = \left((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right)' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ *($\mathbf{X}'\mathbf{X}$, hence its inverse, is symmetric)*:

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\sigma^2\mathbf{I}\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

σ^2 is a scalar, so it can be pulled out front.

Theorem 38 Sampling Distribution of $\hat{\beta}$.
By linearity of normal random variables,

$$\boxed{\hat{\boldsymbol{\beta}} \sim N_{p+1}(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1})}$$

So when the assumptions hold, the estimators are

- **unbiased** (on average estimate the value they should),
- generally **correlated** (off-diagonals of $(\mathbf{X}'\mathbf{X})^{-1}$ not guaranteed to be 0),
- dependent on the same constant variance σ^2 as the errors.

Violations Break the Sampling Distribution

- **Linearity**: the mean of $\hat{\boldsymbol{\beta}}$ is no longer $\boldsymbol{\beta} \Rightarrow$ estimates are biased.
- **Constant variance**: no longer a single σ^2 in the variance.
- **Uncorrelated errors**: the variance of the estimators will be under- or over-estimated.
- **Normality**: estimators won't have a normal sampling distribution.

When assumptions are violated, we don't know the correct way to describe variation due to sampling $\Rightarrow$ **inference will be incorrect or misleading**.

Gauss–Markov Theorem

Theorem 39 Gauss–Markov (Gauss 1821, Markov 1900).
Let $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ *(linearity)*, $\mathbf{X}$ full rank, and $\mathrm{Var}(\mathbf{Y}) = \sigma^2\mathbf{I}$ *(constant variance, uncorrelated errors)*. Then the OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ is the **Best Linear Unbiased Estimator (BLUE)** of $\boldsymbol{\beta}$: for any $\mathbf{a}'\boldsymbol{\beta}$, among estimators that are linear in $\mathbf{Y}$ and unbiased, $\mathbf{a}'\hat{\boldsymbol{\beta}}$ has **minimum variance**. *OLS is the "best" — lowest sampling variance in the class of linear unbiased estimators.*

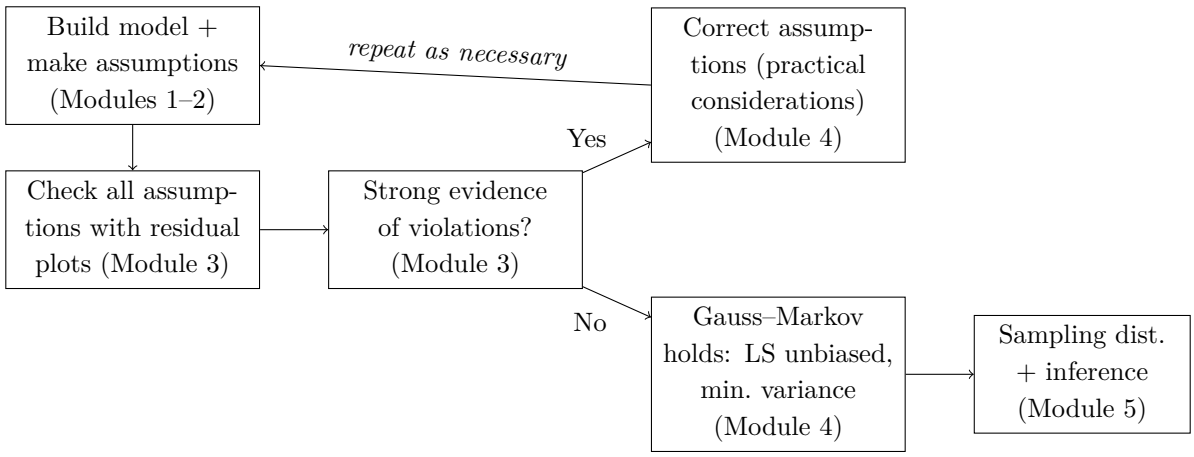
Ordinary Least Squares	**Maximum Likelihood**
Minimizes $RSS = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \sum_{i=1}^{n} \hat{e}_i^2$	Maximizes $\ln L(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{Y}, \mathbf{X}) \propto -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i\boldsymbol{\beta})^2$ *(the sum is the RSS)*
$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$	$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$
Requires: linearity, constant variance, uncorrelated errors. **No** normality assumption.	Requires: linearity, constant variance, uncorrelated errors, **and normality**.

Which Estimation Procedure to Use?

- If normality does not hold, OLS still gives an unbiased and most precise estimator of $\boldsymbol{\beta}$.
- If normality holds, MLE and OLS are equivalent (and both BLUE).
- OLS is **always** more precise than any other estimator linear in $\mathbf{Y}$.
- If inference (hypothesis tests, CIs) isn't important, OLS is the better choice — but the point of statistics is to do inference.

Ideally, normality is satisfied so OLS and MLE are the same.

Summary



Approaches to Transformations

- **Variance stabilizing transformation**

  - **Only** on Y; corrects non-constant variance (can improve other violations, but can also make them worse).
  - No obvious choice; common: $\ln(y-c)$, $\sqrt{y-c}$ (variance usually increases with Y).
  - Use judgement from the plots and the funnel direction (e.g. $\ln(c-y)$).
- **Box–Cox transformation** (on Y and/or X)
  - Corrects non-normality in Y and/or X (sometimes helps non-linearity).
  - Estimate Y^{λ} and/or X^{λ} by maximum likelihood.
  - Guide for simpler transformations: $\hat{\lambda}=0 \to \ln(\cdot)$; $\hat{\lambda}\neq 0 \to Y^{\hat{\lambda}}$ or $X^{\hat{\lambda}}$.

  Generally: transform X for linearity issues.
- We **cannot** correct correlated errors after data collection — it **must** be addressed at the study design / data collection stage.

Workflow

1. Plot data: scatterplot, scatterplot matrix (pairs plot), histogram, ...
2. Plot residuals: all residual plots, QQ plot, ...
3. Fix the **worst** violation first with an appropriate transformation.
4. Check residual plots again after *every* transformation.
5. Avoid complex/too many transformations — keep things interpretable, and keep in mind what variables represent.

Weisberg Ch 8 has a good discussion on transformations.

Remark (Module Takeaways).
- Why does a variance stabilizing transformation correct non-constant variance?
- What is a Box–Cox transformation trying to achieve?
- How do we select and implement transformations to correct model violations?
- What does this change in how we interpret model coefficients?
- Why is it important to attempt to correct model violations?
- Why is the OLS estimator the "best" estimator?

Next week: inference on coefficients and predicted values. The first hour of Module 5 (inference on coefficients) is the last material on the term test.

Week 5: Inference on Linear Regression Components

Remark (Where We Are). Build model + make assumptions → check all assumptions with residual plots → strong evidence of violations? If **yes**, correct the assumptions and repeat. If **no**, Gauss–Markov holds (LS estimates are unbiased with minimum variance) ⇒ **today**: use this to build sampling distributions and make inference.

Module Outline

- Sampling distribution of coefficients
- Inference on coefficients: confidence intervals and hypothesis tests
- Inference for the mean response
- Predictions for actual responses

Sampling Distribution of $\hat{\beta}$

Remark (Why Sampling Distributions).

- When we estimate anything, we need to quantify its error: how much will $\hat{\boldsymbol{\beta}}$ change from sample to sample?
- Defining a sampling distribution uses **assumptions about the population**. If the assumptions don't hold, the sampling distribution is wrong.
- The sampling distribution is our reference for what is considered normal variation to expect; we compare estimates to it to decide whether a value is reasonable.

Definition - Pivotal Quantity

Always of the form

$$\boxed{\frac{\text{Estimator} - \text{Truth}}{\text{Standard Error}}} \qquad \text{e.g.} \qquad \frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)}$$

It is the basis of confidence intervals and hypothesis tests. *E.g. the z-score of the sample mean,* $z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ *(STA238/220).*

Theorem 40 Sampling Distribution of $\hat{\beta}$ (from Week 4).
From the assumptions $\mathbf{Y} \mid \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, by properties of expectation, variance and linearity of the normal,

$$\boxed{\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1})}$$

So when the assumptions hold, the estimators are

- **unbiased** (on average estimate the value they should),
- generally **correlated** (off-diagonals of $(\mathbf{X}'\mathbf{X})^{-1}$ not guaranteed to be 0),
- dependent on the same constant variance σ^2 as the errors.

Definition - Covariance Matrix of $\hat{\beta}$

For MLR:

$$\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \begin{pmatrix} \text{Var}(\hat{\beta}_0 \mid \mathbf{X}) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid \mathbf{X}) & \cdots & \text{Cov}(\hat{\beta}_0, \hat{\beta}_p \mid \mathbf{X}) \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid \mathbf{X}) & \text{Var}(\hat{\beta}_1 \mid \mathbf{X}) & \cdots & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p \mid \mathbf{X}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_p \mid \mathbf{X}) & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p \mid \mathbf{X}) & \cdots & \text{Var}(\hat{\beta}_p \mid \mathbf{X}) \end{pmatrix}$$

For SLR:

$$\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \frac{\sigma^2}{\sum(x_i - \bar{x})^2} \begin{pmatrix} \sum x_i^2/n & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix}$$

The estimated coefficients have non-zero covariance.

Remark (Using the Covariance Matrix). Recall $\text{Var}(\mathbf{A}'\mathbf{Y} + \mathbf{a}) = \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A}$ if $\text{Var}(\mathbf{Y}) = \boldsymbol{\Sigma}$. We will often need the variance of expressions such as $\mathbf{x}_0'\hat{\boldsymbol{\beta}}$ for inference, interpretation and prediction (later this module).

Exercise 41 Variance of the Residuals.
Show $\text{Var}(\hat{\epsilon}_i \mid \mathbf{X}) = \sigma^2(1-h_{ii})$ (Week 3), where h_{ii} is the i-th diagonal element of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.
Hints: work with the vector $\hat{\boldsymbol{\epsilon}} = \mathbf{Y} - \hat{\mathbf{Y}}$*; write* $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$*,* $\hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$*;* $\mathbf{H}^2 = \mathbf{H}$ *and* $(\mathbf{I} - \mathbf{H})^2 = ?$
$(\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - \mathbf{H}$, and $\mathbf{I} - \mathbf{H}$ is symmetric. With $\mathbf{A}' = \mathbf{I} - \mathbf{H}$,

$$\text{Var}(\hat{\boldsymbol{\epsilon}} \mid \mathbf{X}) = (\mathbf{I} - \mathbf{H})\,\sigma^2\mathbf{I}\,(\mathbf{I} - \mathbf{H})' = \sigma^2(\mathbf{I} - \mathbf{H})^2 = \sigma^2(\mathbf{I} - \mathbf{H}),$$

whose i-th diagonal entry is $\sigma^2(1 - h_{ii})$.

The $\hat{\beta}_j$'s Are Correlated

- Generally $\text{Cov}(\hat{\beta}_i, \hat{\beta}_j \mid \mathbf{X}) \neq 0$ — look at the (i, j)-th element of $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.
- The more correlated the predictors X_i and X_j are in the sample, the more correlated **in the opposite direction** $\hat{\beta}_i$ and $\hat{\beta}_j$ are.
- We want to avoid **multicollinearity**: predictors that are [close to] linearly dependent. Perfect multicollinearity makes $\mathbf{X}'\mathbf{X}$ not invertible.

Example 42 Correlated $\hat{\beta}$'s: House Prices
True model: $Y = 1 + 3X_1 + 2X_2$ (in \$100,000), $X_1 = \#$ bedrooms, $X_2 = \#$ bathrooms. The dataset is almost exclusively condos with the same number of bedrooms and bathrooms: $X_1 \approx X_2$, so $\text{corr}(X_1, X_2) \approx 1$.

- $\hat{Y} = 1 + 3X_1 + 2X_2$, $\hat{Y} = 1 + 4X_1 + 1X_2$, $\hat{Y} = 1 + 5X_1 + 0X_2$, $\hat{Y} = 1 + 6X_1 - 1X_2$, … are all likely and valid on the training data $\Rightarrow \text{corr}(\hat{\beta}_1, \hat{\beta}_2) \approx -1$.
- The more positively correlated X_1, X_2 are, the more **negatively** correlated $\hat{\beta}_1$, $\hat{\beta}_2$ are, **and** the higher the variance of $\hat{\beta}_1$ and $\hat{\beta}_2$.

Next week (feature selection): we probably don't need both X_1 *and* X_2.

Unknown σ^2 and the t Distribution

Definition - Estimating σ^2 (Week 3)

σ^2 is an unknown population parameter; estimate it with

$$\boxed{s^2 = \hat{\sigma}^2 = \underbrace{\frac{1}{n-(p+1)}}_{1/df} \underbrace{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}_{RSS} = \frac{\hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}}}{n-(p+1)} = \frac{RSS}{df}}$$

where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \cdots$. So we use $s^2(\mathbf{X}'\mathbf{X})^{-1}$ instead of $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ as $\text{Var}(\hat{\boldsymbol{\beta}})$ — but this **adds additional sampling variation**. *Intercept-only model $y = \beta_0 + \epsilon$ gives $\hat{\beta}_0 = \bar{y}$: this is exactly inference on the mean from last year.*

Remark (Analogy with the Sample Mean). If $y_i \overset{iid}{\sim} N(\mu, \sigma^2)$ then $\bar{y} \sim N(\mu, \sigma^2/n)$, i.e. $\frac{\bar{y}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$. But in practice we used $\frac{\bar{y}-\mu}{s/\sqrt{n}} \sim T_{n-1}$. *Recall* $T = \dfrac{N(0,1)}{\sqrt{\chi^2_{df}/df}}$.

Theorem 43 Pivotal Quantity for $\hat{\boldsymbol{\beta}}$.
Since $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$,

$$\frac{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}}} \sim N(0,1) \qquad \text{and} \qquad \boxed{\frac{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}}{\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}}} \sim T_{n-p-1}}$$

We use this to build confidence intervals and perform hypothesis tests. *Non-standard notation: the square root is* elementwise *(taken on the diagonal), since we think about each $\hat{\beta}_j$ separately.*

Confidence Intervals and Hypothesis Tests for β_j

Remark (General Form). Many inferential procedures take a common form:

- **CIs**: **estimate** $\pm$ (critical value)$\times$(standard error)
- **Test statistics**: $\dfrac{\text{estimate} - \text{truth}}{\text{standard error}}$ *("truth" = the null value)*

The sampling distribution has all of these components:

$$\frac{\overbrace{\hat{\boldsymbol{\beta}}}^{\text{estimate}} - \boldsymbol{\beta}}{\underbrace{\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}}}_{\text{standard error}}} \sim T_{n-p-1} \qquad \text{cf.} \quad \frac{\bar{y}-\mu}{s/\sqrt{n}} \sim T_{n-1}$$
$$\text{critical value} \uparrow$$

Definition - $100(1-\alpha)\%$ Confidence Interval for β_j

$$\hat{\beta}_j \pm t_{\alpha/2,\, n-p-1}\; s\sqrt{(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}}$$

- α is the chosen significance level; $1-\alpha$ is the confidence level.
- The **critical value** is the $\alpha/2$ quantile of the T distribution with $n-p-1$ degrees of freedom. *In R:* `qt(alpha/2, df)` *for the negative critical value,* `qt(1-alpha/2, df)` *for the positive one.*
- $(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}$ is the $(j+1)$-th element of the main diagonal, e.g. $\text{Var}(\hat{\beta}_0) = s^2(\mathbf{X}'\mathbf{X})^{-1}_{(1,1)}$.

Compare $\bar{y} \pm t_{\alpha/2,n-1}\, s/\sqrt{n}$.

Confidence Interval Interpretation

- Built using the data and the sampling distribution.
- A window $2 \times t_{\alpha/2,n-p-1}$ standard errors wide, centred on the estimate $\hat{\beta}_j$. We would see the true β_j in this window $(1-\alpha)\%$ of the time if repeated samples are taken.
- **Interpretation**: $(1-\alpha)\%$ of all intervals computed using data repeatedly obtained from the same population would contain the true β_j. *Can also say: plausible values of* β_j *with* $(1-\alpha)\%$ *confidence.*

Definition - Hypothesis Test of $H_0 : \beta_j = \beta_j^0$

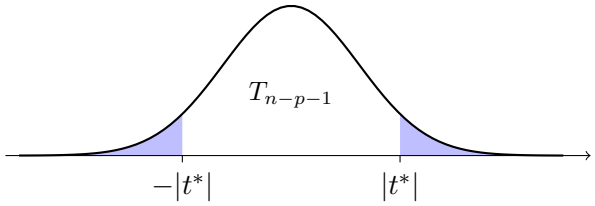
Test statistic:

$$t^* = \frac{\hat{\beta}_j - \beta_j^0}{s\sqrt{(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}}} = \frac{\text{estimate} - \text{null value}}{SE(\text{estimate})} \qquad \text{cf.} \quad \frac{\bar{y}-\mu_0}{s/\sqrt{n}}$$

- β_j^0 is the **null value**: usually 0 for regression, but can be anything. Most common: $H_0 : \beta_j = 0$ vs. $H_A : \beta_j \neq 0$ — "do I need X_j in the model?"
- The standard error $s\sqrt{(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}} = SE(\hat{\beta}_j)$ standardizes the difference between the estimate and the hypothesized value.
- Compare to the expected variability under T_{n-p-1} to see whether the hypothesized value is plausible.

Hypothesis Test Interpretation

- $H_0 : \beta_j = 0$ tests the null that **no linear relationship exists between X_j and Y** (in the presence of the other predictors).
- Under H_0, $t^* \sim T_{n-p-1}$. Conclude by comparing to the sampling distribution:
  - If $|t^*| > t_{\alpha/2,n-p-1}$, reject H_0.
  - If $P(|T_{n-p-1}| \geq |t^*|) < \alpha$, reject H_0. *In R:* `2*pt(-abs(tstar), df)` *(both tails).*
- If rejected: claim a significant linear relationship exists between X_j and Y in the presence of the other predictors.



Two-sided p-value: shaded area in both tails.

Remark (CIs and Tests Agree). The usual relationship applies: if a 95% CI does not contain the hypothesized value, the null hypothesis would be rejected at the 0.05 significance level.

Example: Room Cleaning Dataset

Example 44 Setup
Data from Foster, Stine, and Waterman (1997, p. 9): 53 observations. Model the number of rooms cleaned (Y) by the size of the cleaning crew (X). *From the scatterplot, constant variance is mildly violated (spread grows with crew size).*
Step 0: read in data, build the design matrix and response vector (so we can also compute manually):

```
data <- read.table("cleaning.txt", header=T)
Y <- data[,3]                                  # Rooms Cleaned
X <- as.matrix(cbind(rep(1, 53), data[,2]))    # Predictor: crew size
```

Example 45 Built-in R: `summary()`
```
model <- lm(Rooms ~ Crews, data=data)
summary(model)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.7847     2.0965   0.851    0.399
Crews         3.7009     0.2118  17.472   <2e-16 ***
Residual standard error: 7.336 on 51 degrees of freedom
Multiple R-squared: 0.8569, Adjusted R-squared: 0.854
F-statistic: 305.3 on 1 and 51 DF, p-value: < 2.2e-16
```

- `Estimate`: $\hat{\beta}_j$; `Std. Error`: $SE(\hat{\beta}_j) = \sqrt{\text{Var}(\hat{\beta}_j)}$
- `t value`: test statistic $t^* = \hat{\beta}_j / SE(\hat{\beta}_j)$
- `Pr(>|t|)`: p-value for $H_0 : \beta_j = 0$ vs. $H_A : \beta_j \neq 0$
- `Residual standard error`: s; degrees of freedom $= n - p - 1 = 53 - 2 = 51$

Example 46 CIs from the Summary Output
CIs are always **estimate** $\pm$ (critical value)$\times$(standard error), with estimate $= \hat{\beta}_j$, critical value = `qt(alpha/2, df)` or `qt(1-alpha/2, df)`, SE $= SE(\hat{\beta}_j)$:

```
summ  <- summary(model)
```

```
coefs <- summ$coefficients   # the 4 columns of summary output
U <- coefs[,1] + qt(0.975, df = 51)*coefs[,2]   # upper bound
L <- coefs[,1] - qt(0.975, df = 51)*coefs[,2]   # lower bound
cbind(L, U)
#                    L        U
# (Intercept) -2.424160 5.993558
# Crews        3.275653 4.126134
```

Example 47 CIs with `confint()`
Defaults to 95% CIs; works with many model objects (GLMs, nonlinear/weighted least squares, ...); can specify which parameters (default: all).

```
confint(model)                  # 95%
confint(model, level=0.9)       # 90%: Crews [3.346039, 4.055748]
confint(model, parm = "Crews")  # only beta1
confint(model, parm = 1)        # by order; 1 is the intercept
```

Remark (Term Test Coverage). **Term test coverage ends here.**

Study Tips: Transformations and Assumption Checking

- Practice with different datasets.
- Generate datasets from distributions that violate the assumptions and check residual plots to build intuition.
- Try different transformations: which violations get fixed? Which get worse?
- Generate datasets that *satisfy* the assumptions and look at their residual plots — what does an acceptable amount of noise/deviation look like?

Rough Guidelines for Transformations

- No universal checklist; it varies from problem to problem.
- Issues with **linearity**: consider transforming the **predictors**.
- Issues with **residual variance/distribution**: consider transforming the **response** to stabilize the variance and/or fix the residual distribution.

Example 48 Room Cleaning: Manual Calculations
Step 1: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ gives the fitted model $\hat{y}_i = 1.785 + 3.701x_i$.

```
xtx_inv  <- solve(t(X) %*% X)   # (X'X)^-1
xty      <- t(X) %*% Y          # X'Y
beta_hat <- xtx_inv %*% xty     # 1.784699, 3.700893
```

Step 2: RSS and s^2:

$$RSS = 2744.796, \qquad s^2 = \frac{RSS}{n-p-1} = \frac{2744.796}{53-1-1} = 53.82$$

```
Y_hat <- X %*% beta_hat
RSS   <- sum((Y_hat - Y)^2)      # 2744.796
RSS/51                           # 53.81953
```

Step 3: variances. `xtx_inv` $= \begin{pmatrix} 0.081666 & -0.007235 \\ -0.007235 & 0.000834 \end{pmatrix}$, so

$$\text{Var}(\hat{\beta}_0) = s^2(\mathbf{X}'\mathbf{X})^{-1}_{(1,1)} = 53.82(0.08167) = 4.395$$
$$\text{Var}(\hat{\beta}_1) = s^2(\mathbf{X}'\mathbf{X})^{-1}_{(2,2)} = 53.82(0.00083) = 0.0449$$

`s2*xtx_inv` also gives $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -0.3894$.

Step 4: critical value and distribution. Sampling distribution is $T_{n-p-1} = T_{51}$, $\alpha = 0.05$: $|t_{0.025,51}| = 2.00$ *(`qt(0.025, 51)` $= -2.007584$)*.

Step 5: confidence interval:

$$\hat{\beta}_0 \pm t_{0.025,51}\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}_{(1,1)}} = 1.785 \pm 2.00\sqrt{4.395} = [-2.42,\ 5.99]$$

Interpretation: we are 95% confident that β_0 is between -2.42 and 5.99. *If we repeat this process with repeated samples, 95% of the created CIs will contain the true β_0.*

```
beta_hat[1] + crit_val*sqrt((s2*xtx_inv)[1,1])   # -2.42416
beta_hat[1] - crit_val*sqrt((s2*xtx_inv)[1,1])   #  5.993558
```

Step 5 (test): in linear regression we almost always test $H_0 : \beta_j = 0$ vs. $H_A : \beta_j \neq 0$.

$$H_0 : \beta_1 = 0 : \qquad t^* = \frac{\hat{\beta}_1 - 0}{\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}_{(2,2)}}} = \frac{3.701}{\sqrt{0.0449}} = 17.47$$

Either compute the p-value or compare with the critical value: $17.47 \gg 2.00 \Rightarrow$ **reject** H_0; a statistically significant linear relationship exists.

```
beta_hat[2]/sqrt((s2*xtx_inv)[2,2])   # 17.47211
2*pt(-17.47, 51)                      # 3.573044e-23
```

Exercise 49 Repeat for β_1 (CI) and β_0 (Test).
CI for β_1: $3.701 \pm 2.0076\sqrt{0.04487} = 3.701 \pm 2.0076(0.2118) =$ **[3.276, 4.126]**, matching `confint()`.
Test for β_0: $t^* = 1.785/\sqrt{4.395} = 1.785/2.0965 = 0.851 < 2.00$, p-value $= 0.399 > 0.05 \Rightarrow$ **fail to reject** $H_0 : \beta_0 = 0$ (consistent with the CI $[-2.42, 5.99]$ containing 0).

Exercise 50 Without Matrices.
Using the data file, repeat the example with the SLR formulas, recalling

$$\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\begin{pmatrix} \sum x_i^2/n & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix},$$

i.e. $\text{Var}(\hat{\beta}_1) = \dfrac{s^2}{SS_{xx}}$ and $\text{Var}(\hat{\beta}_0) = s^2\dfrac{\sum x_i^2}{n\,SS_{xx}} = s^2\Big(\dfrac{1}{n} + \dfrac{\bar{x}^2}{SS_{xx}}\Big)$.

Inference for the Mean Response $E(\mathbf{Y} \mid \mathbf{X})$

Three Kinds of Inference with Linear Regression

- On $\boldsymbol{\beta}$: the relationship between the response and predictors *($\boldsymbol{\beta}$ are unknown population parameters)*.
- On $E(\mathbf{Y} \mid \mathbf{X})$, estimated by $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ *(an unknown population parameter, similar to inference on μ previously)*.
- **Prediction** about an individual observation's y_0 *(y_0 is an unknown future value)*.

Definition - Estimated Mean Response

For each $\mathbf{x}_0' = (1, x_1, x_2, \ldots, x_p)$, the estimate is

$$\hat{y}_0 = \hat{E}(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0') = \mathbf{x}_0'\hat{\boldsymbol{\beta}} = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

This is a **linear combination of $\mathbf{Y}$**, so we can use $\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n)$ and $\mathbf{A}'\mathbf{Y} \sim N(\mathbf{A}'\boldsymbol{\mu}, \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A})$ with $\mathbf{A}' = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Proposition 51 Sampling Distribution of $\hat{y}_0$.

Mean:

$$E(\hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0) = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathbf{X}\boldsymbol{\beta} = \mathbf{x}_0'\boldsymbol{\beta}$$

Variance, with $\mathbf{A} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$:

$$\mathrm{Var}(\hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0) = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\sigma^2\mathbf{I}\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$$

Equivalently $\mathrm{Var}(\mathbf{x}_0'\hat{\boldsymbol{\beta}}) = \mathbf{x}_0'\,\mathrm{Var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0$. Hence

$$\boxed{\hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0 \sim N\big(\mathbf{x}_0'\boldsymbol{\beta},\ \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0\big)}$$

Same issue as before: σ^2 is unknown, so estimate with s^2, which adds sampling variability $\Rightarrow T$ distribution:

$$\boxed{\frac{\hat{y}_0 - \mathbf{x}_0'\boldsymbol{\beta}}{\sqrt{s^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim T_{n-p-1}}$$

Definition - $(1-\alpha)\%$ CI for the Mean Response $\mathbf{x}_0'\boldsymbol{\beta}$

$$\boxed{\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2,\,n-p-1}\ s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}}$$

Same T_{n-p-1} distribution as before; a window $2\,t_{\alpha/2,n-p-1}$ standard errors wide, centred on the estimate $\mathbf{x}_0'\hat{\boldsymbol{\beta}}$.

For SLR the standard error simplifies to

$$\sqrt{\mathrm{Var}(\hat{y}_0 \mid X, x_0)} = s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}}$$

Exercise 52 SLR Standard Error from the Matrix Formula.
With $\mathbf{x}_0 = (1, x_0)'$ and $SS_{xx} = \sum(x_i - \bar{x})^2$,

$$\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \frac{1}{SS_{xx}}\Big(\frac{\sum x_i^2}{n} - 2x_0\bar{x} + x_0^2\Big) = \frac{1}{SS_{xx}}\Big(\underbrace{\frac{\sum x_i^2}{n} - \bar{x}^2}_{SS_{xx}/n} + (x_0 - \bar{x})^2\Big)$$

$$= \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{SS_{xx}}$$

The CI is narrowest at $x_0 = \bar{x}$ and widens as x_0 moves away from it.

Definition - Hypothesis Test for the Mean Response

For $H_0 : y_0 = y_0^0$,

$$t^* = \frac{\hat{y}_0 - y_0^0}{s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim T_{n-p-1} \text{ under } H_0$$

Think of the mean response as a parameter similar to μ (the conditional population mean for individuals with attributes $\mathbf{x}_0$). Reject H_0 if $|t^*| > t_{\alpha/2,n-p-1}$, or if $P(|T_{n-p-1}| \geq |t^*|) < \alpha$. *Not really done in practice — usually we create confidence intervals.*

Example 53 Room Cleaning: CI for Mean Response
What is the average number of rooms cleaned with a cleaning crew of size 5? Fitted model $\hat{y}_i = 1.785 + 3.701x_i$.

```
new <- data.frame(Crews=5)   # name must match the original data
predict(model, newdata=new, interval="confidence", level=0.95)
#        fit      lwr      upr
# 1 20.28917 17.73171 22.84662

new <- data.frame(Crews=c(4,2,5))   # several points at once
predict(model, newdata=new, interval="confidence", level=0.95)
#         fit       lwr      upr
# 1 16.588272 13.750660 19.42588
# 2  9.186485  5.699374 12.67360
# 3 20.289166 17.731710 22.84662
```

Exercise 54 Verify the CI Manually.
Use R to find $(\mathbf{X}'\mathbf{X})^{-1}$, $|t_{0.025,51}|$, SS_{xx}, SS_{yy}, SP_{xy}, $\bar{X}$, $\bar{Y}$, then complete the calculation with both the SLR and MLR formulas.
MLR: with $\mathbf{x}_0 = (1, 5)'$, $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.081666 - 2(5)(0.007235) + 25(0.000834) = 0.03015$, so

$$SE = \sqrt{53.82(0.03015)} = \sqrt{1.623} = 1.274, \qquad 20.289 \pm 2.0076(1.274) = \mathbf{[17.73,\ 22.85]}.$$

Prediction Interval for an Actual Response y_0

Remark (Actual vs. Mean Response).

- The regression model gives predicted values **only** for $E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0)$ — estimates of conditional mean responses. We cannot exactly predict any individual response. $E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0)$ is a fixed but unknown population parameter.
- But an important goal of regression is **prediction**: for an individual with known $\mathbf{x}_0$, what is their y_0?
- The actual response y_0 is a **realization of a random variable** $Y \mid \mathbf{X} = \mathbf{x}_0$: it can take any possible value, and is typically not $E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0)$ (y_0 is not necessarily on the line/plane of best fit). So we must account for the random error as well.

Definition - Prediction Error

Using $\hat{y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$ as the predicted value,

$$y_0 - \hat{y}_0 = \mathbf{x}_0'\boldsymbol{\beta} + \varepsilon_0 - \hat{y}_0 = \underbrace{(\mathbf{x}_0'\boldsymbol{\beta} - \hat{y}_0)}_{\text{fit of mean}} + \underbrace{\varepsilon_0}_{\text{random error}}$$

Two sources of error:

- the random error in the population, ε_0;
- how well the model fits the mean response, $\mathbf{x}_0'\boldsymbol{\beta} - \hat{y}_0 = \mathbf{x}_0'\boldsymbol{\beta} - \mathbf{x}_0'\hat{\boldsymbol{\beta}}$.

Proposition 55 Distribution of $y_0 - \hat{y}_0$.
Assuming the model assumptions hold, $\varepsilon_0 \sim N(0, \sigma^2)$ for the new observation is independent of the sample $\mathbf{Y}$, hence of $\hat{y}_0$. Mean:

$$E(y_0 - \hat{y}_0) = \mathbf{x}_0'\boldsymbol{\beta} + 0 - \mathbf{x}_0'\boldsymbol{\beta} = 0$$

Variance (no covariance term, by independence):

$$\mathrm{Var}(y_0 - \hat{y}_0) = \mathrm{Var}(\varepsilon_0) + \mathrm{Var}(\hat{y}_0) = \sigma^2 + \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$$

A linear combination of normals is normal, so

$$y_0 - \hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0 \sim N\big(0,\ \sigma^2[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]\big)$$

Estimating σ^2 with s^2 adds sampling variability $\Rightarrow T_{n-p-1}$:

$$\frac{y_0 - \hat{y}_0}{\sqrt{s^2[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]}} \sim T_{n-p-1}$$

Notice the extra "+1" term in the variance (from ε_0).

Definition - $(1-\alpha)\%$ Prediction Interval for y_0

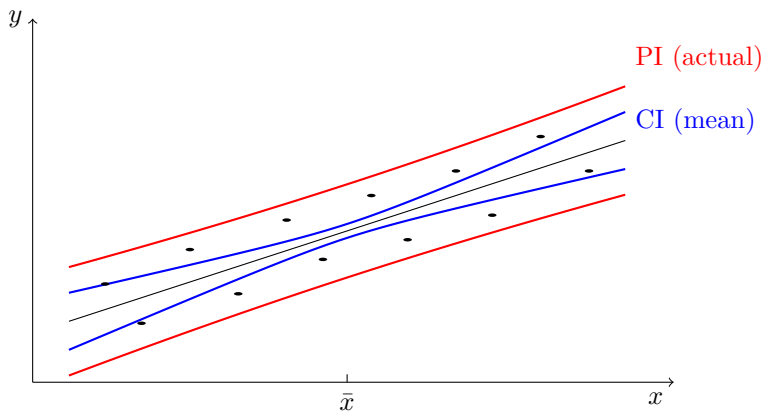
$$\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2,\,n-p-1}\; s\sqrt{1+\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$$

- We cannot predict the exact value of y_0 since we do not know ϵ_0; instead give a range of possible values for the actual response.
- **Not** a confidence interval: we are not estimating an unknown parameter — y_0 is a realization of a random variable.
- Same T_{n-p-1} distribution; window $2\,t_{\alpha/2,n-p-1}$ standard errors wide, centred on the estimated mean response $\hat{y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$.
- **Wider** than the CI for the mean response $E(\mathbf{Y} \mid \mathbf{X}, \mathbf{x}_0)$.

For SLR:

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{\alpha/2,\,n-2}\; s\sqrt{1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{\sum(x_i-\bar{x})^2}}$$

Follows from the previous exercise by adding 1 inside the root.



Both bands are centred on the fitted line and narrowest at $\bar{x}$; the PI is always wider.

Example 56 Room Cleaning: PI for Actual Response

How many rooms will this cleaning crew of size 5 clean?

```
new <- data.frame(Crews=5)
predict(model, newdata=new, interval="prediction", level=0.95)
#        fit      lwr      upr
# 1 20.28917 5.340774 35.23756

new <- data.frame(Crews=c(4,2,5))
predict(model, newdata=new, interval="prediction", level=0.95)
#         fit       lwr      upr
# 1 16.588272  1.589410 31.58713
```

```
# 2  9.186485 -5.948698 24.32167
# 3 20.289166  5.340774 35.23756
```

The prediction intervals are **wider** than the confidence intervals, but the **centres are the same**.

Exercise 57 Verify the PI Manually.
Use R to find $(\mathbf{X}'\mathbf{X})^{-1}$, $|t_{0.025,51}|$, SS_{xx}, SS_{yy}, SP_{xy}, $\bar{X}$, $\bar{Y}$; do the rest by hand with both the SLR and MLR formulas.
Reusing $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.03015$ at $x_0 = 5$:

$$SE = \sqrt{53.82(1 + 0.03015)} = \sqrt{55.44} = 7.446, \qquad 20.289 \pm 2.0076(7.446) = \mathbf{[5.34,\ 35.24]}.$$

Confidence vs. Prediction Intervals

Confidence interval	**Prediction interval**
For the **mean response**	For a **future/specific observation**
"What would be the average selling price for a house with characteristics $\mathbf{x}_0$?"	"What is the predicted sale price for *this* house with characteristics $\mathbf{x}_0$?"
SE: $s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$	SE: $s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$

- Prediction intervals are **always wider** than confidence intervals *(think: variability of y_i vs. $\bar{y}$)* — they include variation in estimating the conditional mean *and* variation around the conditional mean.
- Both are centred at $\mathbf{x}_0'\hat{\boldsymbol{\beta}}$.
- Some sources call both "confidence intervals" but specify whether the interest is a single future response or the mean response.

Remark (Module Takeaways).
- How did we determine the properties of the sampling distribution, and where did the assumptions play a role?
- Why do we use a T distribution with the sampling distribution in practice?
- How do we compute confidence/prediction intervals and conduct hypothesis tests on regression components?
- How are the inferential procedures concluded?
- What is the difference between estimating a mean response and predicting an actual response?

Next week: variance decomposition — how much of the total variation is explained by the predictors?